Technology in Society 82 (2025) 102955



Contents lists available at ScienceDirect

Technology in Society

journal homepage: www.elsevier.com/locate/techsoc





Virtual-physical alliance in the metaverse ecosystem: Business model evolution and optimal deployment of knowledge management strategy

Baogui Xin [a,*], Yaping Song [a], Wei Peng [a], Hui Tan [b]

[a] College of Economics and Management, Shandong University of Science and Technology, China
[b] School of Business and Management, Royal Holloway, University of London, UK

ARTICLE INFO

Keywords:
Knowledge management
Virtual-physical alliance
Game theory
Business models
Metaverse ecosystems

ABSTRACT

With information technology development and cross-organizational strategic collaboration, implementation scenarios and scope of knowledge management strategies have become more extensive. Leveraging advanced knowledge management approaches to seek innovation and ensure sustainability has become imperative. The metaverse promotes the innovation and evolution of business models, achieving a business reconstruction from traditional e-commerce models to virtual-physical alliances, and the knowledge management strategy from different market forces accelerates this process. This study explores the internal logic behind the emergence of new business models in the metaverse, providing a robust framework for understanding the inseparable relationships between metaverse technology, knowledge management, and business model innovation. By analyzing the synergistic effects of metaverse technology-driven brand strategy interaction, market value acquisition, and knowledge management deployment, this research elucidates how brands can optimize customer experiences while maintaining sustainable growth. Utilizing quantitative data analysis and qualitative case study, the study identifies key mechanisms by which the metaverse enhances product differentiation, the fusion of virtual and physical economies, and sustainable marketing strategies through knowledge-based business model innovation. The findings suggest robust knowledge management frameworks in metaverse ecosystems significantly improve customer satisfaction, higher conversion rates, and sustained profitability. This paper provides actionable insights for metaverse market participants aiming to harness the potential of advanced technologies and knowledge-driven approaches to promote long-term economic sustainability and deliver superior value to customers. The implications of this study extend to policymakers and technology service providers seeking to foster inclusive and resilient digital marketplaces.

1. Introduction

The metaverse has strategic significance for current economic and social development. The adoption of disruptive technologies such as augmented reality, virtual reality, blockchain, and artificial intelligence (AI) has made it possible to apply metaverse business models in the real world (Huynh-The et al., 2023; Kuo & Choi, 2024). The waves of technological innovation in the metaverse attract the joint participation and creation of stakeholders from around the world. With the continuous improvement of technological infrastructure and the generation of various social and economic activities enhanced by technologies, the metaverse has realized larger and more efficient business ecosystems (Tomini et al., 2024; Zabel et al., 2023). The metaverse ecosystems accelerate the innovation and evolution of business models, and the free trading of virtual products measured by digital assets creates conditions for the integration of physical and virtual (Moro-Visconti, 2022; Zhang & Ye, 2023). Many brands are utilizing emerging technologies in the metaverse to improve their retail business processes, advance influence, and expand markets (Bruni et al., 2025; Hadi et al., 2024; Jafar et al., 2023). For example, Adidas launched its own virtual fashion collection and allowed consumers who purchase such digital assets to exchange them for corresponding physical products (Jafar et al., 2023). Pacsun implemented the business model of digital art bundled with physical shopping privileges, offering free shipping and price discounts for physical clothing to consumers (Yoo et al., 2023).

Business model innovation is the process by which enterprises utilize

This article is part of a special issue entitled: Role of Knowledge Management published in Technology in Society.

* Corresponding author.
E-mail addresses: xin@sdust.edu.cn (B. Xin), 202180070003@sdust.edu.cn (Y. Song), pengweisd@sdust.edu.cn (W. Peng), hui.tan@rhul.ac.uk (H. Tan).

https://doi.org/10.1016/j.techsoc.2025.102955
Received 14 December 2024; Received in revised form 13 May 2025; Accepted 25 May 2025
Available online 29 May 2025
0160-791X/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

new ways to acquire and create value, which requires enterprises to effectively manage intangible and tangible resources such as technology and knowledge (Mancuso et al., 2024; Tomini et al., 2024). The existing literature evidence shows that technological innovation and knowledge management, which are necessary strategic capabilities of enterprises, play a crucial role in business model innovation (Ancillai et al., 2023; Sjodin et al., 2020). As a key driving force, the process of value creation and acquisition will evolve synchronously with technological changes (Krotov, 2017). The business model innovation centered on value acquisition requires knowledge management processes to locate key organizational information, acquiring internal and external knowledge (Bresciani et al., 2023; Ul-Durar et al., 2023). Adopting advanced technology is also the key to enhancing the organization's strategic capability to achieve greater scope and speed in knowledge acquisition, dissemination, and application (Hwang & Krackhardt, 2020; Liu et al., 2024). The immersive virtual reality technologies give more possibilities for knowledge management processes, making knowledge generation and sharing include many advantages and characteristics (Abdulmuhsin et al., 2024). The metaverse ecosystems integrate different market forces from global partners, promoting the formation of strategic alliances for exchanging and sharing knowledge resources. For example, NVDA created a metaverse production simulation environment for BMW, enabling it to simulate virtual production of electric vehicles in factories worldwide (Chen, Ye, & Huang, 2023). The metaverse enables cross-regional, real-time collaboration and communication among partners, promoting knowledge transfer and sharing among stakeholders (Chen, Ye, & Huang, 2023; Philsoophian et al., 2022; Zhang & Ye, 2023).

Moreover, metaverse technologies can achieve direct customer knowledge performance by enhancing customer engagement and shopping experience. Real-time interaction with customers can help brands quickly identify knowledge about customer preferences and needs, providing support for organizational operational decisions (Im & Lee, 2023; Jang & Nemeh, 2017; Xu et al., 2023). The acquisition and application of customer market knowledge will create additional customer value for enterprises. The emerging technologies in the metaverse ecosystem improve organizations' knowledge management processes, forming good interactive relationships between strategic alliance partners and between enterprises, customers, and consumers, and further promoting their participation and creation in the metaverse ecosystem. Forming a good collaborative ecosystem and value co-creation network is also a key link of business model innovation, which requires organizations to attach importance to stakeholders' strategic dynamics, including new economic transactions with strategic alliance partners and new market strategies (Yi et al., 2020; Zott & Amit, 2008; Zott et al., 2011). In new relationship networks, all stakeholders participate in the creation, acquisition, and transmission of value in new ways, enabling the reshaping and continuous evolution of business models.

Notably, the business model innovation advanced by stakeholder interaction achieves sustainable growth in organizational knowledge performance, i.e., organizations realize long-term stable economic growth through optimized utilization of knowledge resources. Specifically, there are two explanations. Strategic alliances compensate for the lack of organizational knowledge resources through complementary and integrated structures, reducing risks and promoting stability to achieve sustainability (Zahoor et al., 2024). Besides, the acquisition and dissemination of customer knowledge can create two-way value contributions, including customer needs and their experience of using the enterprise's products and services, thus helping enterprises continuously improve the existing business to meet customers, ensuring economic sustainability (Abbas & Sagsan, 2019; Wijethilake, 2017). The improved knowledge management processes in metaverse ecosystems maintain these two external relationships necessary for enterprises to implement relevant strategies, and good strategic alliance dynamics can also optimize customer engagement to achieve sustainable knowledge performance, thereby helping enterprises to implement more sustainable business strategies in comprehensive knowledge management practices.

The metaverse has a profound impact on today's business landscape. Although many studies have examined the motivations and reasons for the formation of various business models in the metaverse, they mostly focus on a kind of specific business model, and the intersections between the metaverse and business model innovation still lack a discussion on the internal logic and fundamental level of the business model evolution. Furthermore, the research on the role of the metaverse on knowledge management is still in its infancy, and we unify the research perspectives of existing literature on the interplay of the metaverse, knowledge management, and business model innovation, exploring the deeper ternary relationship between them.

To investigate the sustainable creation of knowledge performance in virtual-physical alliance business models in the metaverse and how to optimize strategic deployment through alliance dynamics, we propose a conceptual framework focusing on the evolution mechanism of metaverse business models, and the framework includes key elements such as metaverse technology ecosystems, knowledge management practices, and stakeholder relationship networks. Based on this framework, this paper constructs a theoretical model. Through game equilibrium analyses and the case study, our research results indicate that the strategic interactions of virtual-physical alliances affect the final sales performance through subtle decision levels such as brand pricing and technology investment, helping brands optimize the deployment of knowledge management strategies. Theoretically, this paper constructs a comprehensive framework from a new perspective of business model innovation. By incorporating brand strategy dynamics and knowledge management practices into our theoretical model, we explore the evolution of business models in the metaverse ecosystem and the optimal deployment of knowledge management strategies, enriching the theoretical basis for research on the intersection of the metaverse, knowledge management, and business model innovation. From a practical perspective, our research enables practitioners and policymakers to properly understand and leverage the critical roles of technological change, knowledge management, and relationship networks on business model innovation, helping enterprises address sustainability challenges. In summary, we extend the theoretical basis of the existing literature and provide support for the practical implementation of corporate strategy.

The remainder of this paper is organized as follows. Section 2 briefly reviews the relevant literature. Section 3 proposes our framework and model description. We present our research methodology in Section 4, including the game equilibrium analysis and case verification. Section 5 discusses the results. Conclusions are provided in Section 6.

2. Literature review

Our study encompasses several interrelated streams of research. There are two groups of research that focus on game theory: platform-dominated Stackelberg game and strategic interactive decision. For research on the interplay in ternary relationships, the most relevant streams to our work are technology, knowledge management, and business model innovation; metaverse and business model innovation and metaverse-improved knowledge management process.

2.1. Game theory

2.1.1. Platform-dominated Stackelberg game

The Stackelberg game is the most commonly used method for solving supply chain alliance strategic decision-making problems (Taleizadeh et al., 2017), and it was developed by H. Von Stackelberg in 1930. Considering that the coordination and pricing decisions of supply chain members are often related to their market power, this method is typically used to resolve the leadership-follower relationship issue in supply chains (Sun & Ji, 2022; Tsunoda & Zennyo, 2021; Zhou et al., 2015).

When market power is defined, large firms in a dominant position have greater initiative and become leaders in market decisions, while follower firms take follow-up actions after observing the leader's decision. In traditional supply chain decisions, the game equilibrium dominated by commercial entities like upstream manufacturers or suppliers is often considered. However, in recent years, many brands with their own retail platforms have gradually exceeded those of upstream members in terms of status and influence, and the dominant powers of supply chains have also moved to downstream retailers (Ghadimi et al., 2013; Huang et al., 2020). In particular, some platforms can act as a virtual market, be closer to consumers, and have stronger bargaining power. Platform-dominated Stackelberg game has received widespread attention from scholars (Tong & Xiao, 2024; Zhou et al., 2015).

2.1.2. Strategic interactive decision

In complex game strategies, parties usually achieve various alliance dynamics to create and obtain more value. At this point, the determination of the final pricing equilibrium is affected by multiple factors, with the key being strategic interaction among game players. It is precisely because of the characteristics of many platforms that can act as marketplaces that, in addition to the traditional retail wholesale price modes, market pricing modes, also known as agency contracts, are more common in platform or virtual market-based game models (Huang et al., 2023; Zhang, Xu, et al., 2022). Pricing modes reflect the specific transaction forms among members and determine the final strategy sets of the game. Game players reach strategic cooperation through the revenue-sharing contract, which is a way of strategic coordination that can solve decentralized problems and avoid double marginalization (Arani et al., 2016; Xin et al., 2024). Besides product transactions, platforms and their upstream members often establish strategic cooperation on other services, such as technology support (Sun & Ji, 2022), information sharing (Zhang, Wei, & Zhang, 2022), cost-sharing (Liu et al., 2020). This will involve other transfer payment fees beyond the basic pricing modes, and these important strategic interactions and alliance dynamics are key to impacting ultimate decisions.

2.2. Interplay of metaverse, knowledge management, and business model innovation

2.2.1. Technology, knowledge management, and business model innovation

Business model innovation is a systematic effort to create, acquire, and transmit value, which requires enterprises to effectively manage intangible and tangible resources such as knowledge and technology (Mancuso et al., 2024; Tomini et al., 2024). In academic discussions, both technology innovation and knowledge management are crucial for business model innovation. The evolution and improvement of business models require technology as a key driving force, and the activities of value creation and acquisition will evolve synchronously with the continuous advancement of technology (Sjodin et al., 2020). Ancillai et al. (2023) systematically reviewed the impact of digital technologies on business model innovation, noting that digital technologies make the mechanism of creating, acquiring, and delivering value constantly change, thus promoting the digital transformation of business models. Krotov (2017) believed that technological change affects the degree of business model innovation, and when technology is used to provide solutions for improving products and services, business model innovation is progressive, while when digital technology completely changes the mechanism of enterprise value acquisition and transmission, it will be a thorough innovation of business models.

The knowledge-based view holds that knowledge is an important strategic foundation for enterprises and the key to achieving organizational performance (Bresciani et al., 2023; Ul-Durar et al., 2023). Knowledge management is a set of explicit system processes and operational methods used to locate key organizational information and enterprise resources (Liu et al., 2004). Academic literature typically defines knowledge management as a series of knowledge processing activities, including the creation, acquisition, storage, integration, and application of knowledge (Alavi & Leidner, 2001; Giraldo et al., 2019; Wiig, 1997). Enterprises typically resist complex external dynamics by creating, absorbing, sharing, and applying (Bresciani et al., 2023). Adopting advanced technology is also the key to enhancing the organization's strategic capability to achieve greater scope and speed in knowledge acquisition, dissemination, and application (Hwang & Krackhardt, 2020; Liu et al., 2024). Borrowing information technology tools can maximize the utility of knowledge resources and improve the overall benefits of stakeholders (Gu et al., 2021; Xu et al., 2023).

The transformation of business models centered on value acquisition requires effective knowledge management processes to help enterprises create and acquire internal and external knowledge. Just as strong strategic alliances form a complementary strategic structure among partners, business model innovation also requires attention to the dynamics of stakeholders, creating the communication conditions for enterprises to obtain the necessary external resources (Mancuso et al., 2024). Strategic alliances facilitate this process by enhancing business skills with stakeholders, improving the efficiency of organizational management decisions, and ultimately improving performance to achieve organizational goals (Chang, 2003; Scuotto et al., 2022). This strategic cooperation can promote consistency of goals among stakeholders and have a profound impact on existing business models (Di Vaio et al., 2021; Ramon Saura et al., 2023). Knowledge management can achieve consistency of goals among stakeholders through such strategic cooperation and further promote business model innovation through the integration and reorganization of internal and external knowledge (Loorbach & Wijsman, 2013; Martins et al., 2019; von Delft et al., 2019).

Notably, with the continuous development and progress of technology, the business environment and customer preferences are also constantly changing, requiring enterprises to improve products and services in the fiercely competitive environment, thereby enhancing market influence to obtain more customer value (Tomini et al., 2024). Creating customer value through acquiring customer knowledge is also a component of business model innovation, and many business model frameworks attach great importance to customer interests (Buechel & Spinler, 2024; Yoo et al., 2023). Organizations need to master customer knowledge, understand market dynamics to effectively interact with customers, encourage customer participation in the innovation process, and continuously provide knowledge to customers, establishing better customer relationships through two-way knowledge acquisition and transmission (Aghamirian et al., 2015). Effective customer knowledge management is crucial for improving customer satisfaction (Wu et al., 2014), especially with the acquisition of personalized information and data, and it can reshape customized business models and create new sales, product manufacturing, and service processes (Abbas & Khan, 2023; Schniederjans et al., 2020). Consumers engaged in the market can acquire specific knowledge through transactions, guide their purchasing decisions through knowledge sharing and learning, and promote positive consumer behavior (Olan et al., 2024). At this point, the spatial environment characteristics and technologies for providing products and services can affect customer engagement, contributing to customer value creation (Animesh et al., 2011). Enterprises can adopt technology to quickly identify customer preferences to guide decisions, and a well-designed technology experience can also meet consumers' higher purchasing expectations. Therefore, the interplay between technology and knowledge management has become a key element in business model innovation, and they also play an important role in maintaining partner and customer relationships in strategic alliances, thus helping enterprises improve and enhance their ways of value creation, acquisition, and transmission and achieving fundamental changes in business models.

2.2.2. Metaverse and business model innovation

The metaverse has a profound impact, triggering waves of

technological innovation and attracting stakeholders from all over the world to participate and create together. The perfect infrastructure, resources, and relationship network have formed a huge metaverse ecosystem, equipped with various basic elements required for business model transformation, promoting business models to integrate and adjust (Tomini et al., 2024; Zabel et al., 2023). The academic community is increasingly interested in business model innovation under the impact of the metaverse. Business model innovation is the process by which enterprises utilize new ways to acquire and create value. Tomini et al. (2024) found that metaverse technologies can enhance the acquisition, creation, and transmission of internal and external value in business models, which brings new opportunities for business model innovation. Mancuso et al. (2024) explored the case of Meta and Roblox, two metaverse companies, and elaborated on the process of business model innovation centered on value creation and acquisition from the micro level of technology, knowledge, and relationships. Buechel and Spinler (2024) examined the disruptive impact of the metaverse on e-commerce models, providing strategic references for future business dynamics of enterprises.

In the metaverse ecosystem, the new digital pattern will replace existing business strategies, especially traditional retail and e-commerce models. The metaverse can enhance customer engagement and shopping experience, promote the improvement of existing retail industries, and create conditions for their business model innovation (Bruni et al., 2025; Yoo et al., 2023). The metaverse accelerates the evolution and reconstruction of business models in this field, and many brands are seeking opportunities in the metaverse to create value and expand markets, thus building various metaverse business ecosystems that integrate different market forces (Hadi et al., 2024; Jafar et al., 2023). In addition to highlighting a sustainable digital fashion supply chain (Xin et al., 2025), the complementary combination of virtual and physical products is particularly common in metaverse ecosystems. For example, Adidas has constructed virtual-physical exchange purchasing activities where consumers can exchange purchased virtual products for physical products (Jafar et al., 2023). Zhang and Ye (2023) explored the motivation behind the virtual platform bundling sales model and its impact on platform performance, studying the business model reconstruction in the metaverse era. Yoo et al. (2023) examined the business model of digital art bundled with physical shopping privileges, such as offering free shipping and price discounts for physical clothing to consumers. Therefore, the various participation and strategic dynamics of stakeholders such as virtual brands, physical brands, and consumers in the metaverse ecosystem play a crucial role in the integration and restructuring of business models.

2.2.3. *Metaverse-improved knowledge management process*

The strategic significance of the metaverse lies in that it connects and integrates the digital economy and the real economy. The virtual world created by augmented reality, virtual reality, blockchain, artificial intelligence, and other technologies can seamlessly connect physical reality, driving rapid development in industries such as retail, service, hotel tourism, and healthcare (Huynh-The et al., 2023; Zhang & Ye, 2023). Because of these immersive virtual reality technologies, knowledge generation and sharing contain many advantages and characteristics, giving more possibilities to some fields (Abdulmuhsin et al., 2024). The research on the impact of the metaverse on knowledge management processes is still in its infancy, and some scholars have conducted in-depth discussions on specific fields. Abdulmuhsin et al. (2024) studied the application of the metaverse in medical education and pointed out that metaverse technologies can reshape the generation and sharing of existing medical knowledge, providing the impetus for the sustained professional development of medical education. Wang et al. (2022) argued that the metaverse ecosystem can shape new academic research networks and application platforms for education, and based on previous theoretical frameworks, proposed an educational metaverse ecosystem driven by factors such as technology, knowledge, and talent.

In the retail field, the 3D virtual space created by the metaverse enables consumers to have a presence, and it conveys product knowledge to consumers by encouraging them to interact with simulated product content (Jafar et al., 2023). This allows them to register identities through virtual avatars, and the realistic virtual reality interaction experience enhances consumers' perception of product value (Kshetri, 2022; Zhang & Ye, 2023). Based on more innovative technological solutions, these special interactive methods can replace front-line service businesses. The ability to present knowledge more comprehensively and quickly can meet the higher requirements of brands and consumers for knowledge creation, acquisition, and transmission. Besides, virtual avatars allow consumers to match and create independently, incorporating more of their preferences. Enterprises can gain insight into consumers' online behaviors, make personalized recommendations, and create additional customer value through the knowledge and information obtained from this interaction (Hoyer et al., 2020; Lemon & Verhoef, 2016; Sun & Ji, 2022). Other than knowledge generation, acquisition, and transmission within the enterprise, the metaverse ecosystems also shorten the distance between cooperative partners, making various social and economic activities realistic through advanced technologies, thus promoting knowledge transmission and sharing among partners (Chen, Ye, & Huang, 2023; Jen et al., 2020; Philsoophian et al., 2022). Enterprises can integrate different market forces from global cooperative partners by constructing metaverse business ecosystems, enabling virtual face-to-face communication and collaboration among partners around the world across regions and time, which can create a global source of knowledge to help organizations implement knowledge management strategies (Chen & Huang, 2024; Zhang & Ye, 2023).

Under the influence of metaverse technology waves, the strategic dynamics among stakeholders such as producers and consumers involved in creation have formed a vast metaverse ecosystem. It realizes a new strategic relationship of exchange and sharing of tangible and intangible resources. That is, metaverse technologies have formed good interactive relationships between strategic alliance partners and between enterprises and consumers by shaping and improving knowledge management processes, and these relationships further promote these stakeholders' participation in the metaverse ecosystem. Brands and consumers jointly participate in the creation, acquisition, and transmission of value in new ways that are different from existing models, enabling the reshaping and continuous evolution of business models. We unify the above existing literature on the interplay of the metaverse, knowledge management, and business model innovation, revealing their inseparable relationships, and provide a theoretical basis for our research framework and understanding the mechanism of business model evolution.

3. Framework and model description

3.1. *Conceptual framework*

To present the internal logic of business model evolution in the metaverse ecosystem, we propose a conceptual framework to visually depict the ternary relationship between the metaverse, knowledge management, and business model innovation introduced in our theoretical background, which is shown in Fig. 3. The framework explores the specific business models of the metaverse in the fashion field, i.e., the strategic alliance between the virtual brand and physical brand for creating new products and technical services, involving the strategic dynamics between virtual and physical brands and two-way interactions between brands and consumers.

Previous studies have shown that business model innovation is a process by which enterprises create, acquire, and deliver value in new ways through a series of systematic efforts (Mancuso et al., 2024; Tomini et al., 2024). A strategic alliance is an arrangement voluntarily reached between enterprises for the joint development, sharing, or exchange of

technology, products, or services (Gulati, 1998; Zahoor et al., 2024). This process requires building a good collaborative ecosystem and value co-creation network, including conducting new economic transactions with partners and developing new market strategies (Yi et al., 2020; Zott & Amit, 2008; Zott et al., 2011). And the basic elements of technology, knowledge, and relationships constitute the micro-foundation of business model innovation (Mancuso et al., 2024). The emerging technologies in the metaverse have enabled a larger and more efficient business ecosystem, attracting the joint participation and creation of stakeholders (Tomini et al., 2024; Zabel et al., 2023). In this process, a strategic alliance relationship is formed that can exchange tangible and intangible resources, as well as a good interactive relationship between the enterprise, customers, and consumers. Stakeholders can seamlessly communicate knowledge resources such as products and technologies in the new relationship network, including knowledge acquisition, integration, application, and transmission.

Knowledge acquisition is the process of acquiring knowledge from different sources to utilize, and one of the key sources is acquiring the knowledge of customers, including their needs and experience of using the enterprise's products and services, thus helping enterprises continuously improve the existing business to meet customers, ensuring economic sustainability (Abbas & Sagsan, 2019; Wijethilake, 2017). Knowledge integration is the fusion of different professional knowledge from different organizations or individuals to promote cooperation and coordination between enterprises or teams (Grant, 1996; Okhuysen & Eisenhardt, 2002). Enterprises can integrate complementary resources with partners through external network relationships, which can reduce risks and promote stability to achieve economic sustainability (Zahoor et al., 2024). Moreover, integrating knowledge resources can create new businesses and expand global markets (von Delft et al., 2019). Knowledge application refers to the use of knowledge in the process of designing products and services. Therefore, when generating new business processes, knowledge application is required (Mothe et al., 2018). Besides, knowledge transmission is an autonomous activity in which organizations or individuals learn and share knowledge through relational networks (Reagans & McEvily, 2003). Enterprises also need to transfer product knowledge to customers and consumers, present knowledge to them through necessary tools and labels, help them solve problems, obtain good experiences, and enhance their perceived value of products and services (Jafar et al., 2023).

The emerging technologies in the metaverse ecosystem can improve these processes of knowledge management, and complete relationship networks further promote the participation of these stakeholders, overcoming the traditional business model relying on a single one-way knowledge integration for value creation. The new processes of value creation, acquisition, and transmission enable the business model to be reshaped and evolved from the traditional e-commerce model to the retail model under single knowledge integration and ultimately evolved into the virtual-physical alliance model in the metaverse ecosystem. Fig. 1 shows the evolution of business models accompanying the emergence of the metaverse.

In summary, this framework includes four dimensions of knowledge management practice, namely knowledge acquisition, application, transmission, and integration. Knowledge integration refers to the integration of strategic knowledge resources between brands in virtual-physical alliances. The physical brand shares their design knowledge and concepts with the virtual brand that has technology services and showcases the specific design content and details of physical products to it. The virtual brand creates virtual scenes by adopting metaverse technologies and provides specific marketing channel resources for the physical brand. Knowledge acquisition means the ability of brands to acquire knowledge about consumer preferences and market structure in an interactive virtual purchasing environment. Then, through knowledge application, brands can apply the acquired knowledge to jointly develop virtual products belonging to the physical brand style and rely on the acquired market knowledge to help brands design and manufacture products that meet consumer needs. And knowledge transmission is the process in which brands convey product knowledge to consumers and obtain technology experiences when they participate in virtual purchasing. Through two-way interactions with consumers and continuous improvement of business processes, brands can achieve sustainable sales performance in the virtual-physical alliance business model. Based on the framework, our model design will depict strategic interactive decisions in virtual-physical alliances and present the value creation foundation of various knowledge management dimensions under the effect of the metaverse.

3.2. Strategic interactions and game design

We consider a supply chain alliance structure consisting of a physical and a virtual brand, where the physical brand intends to enter the virtual product market through the globally renowned virtual brand or metaverse service provider, with the virtual brand having its own online channel or platform. We design a game dominated by the virtual brand that has its own platform, which is common in previous studies (Sun & Ji, 2022; Tong & Xiao, 2024; Zhou et al., 2015). As a game leader, the virtual brand provides the platform channel or the interactive scene of the metaverse ecosystem for the physical brand to sell products. The physical brand is required to pay an entry commission to the virtual brand as a marketplace regardless of how the brand alliance is implemented, and the product price p is determined by the physical brand itself. We realize a revenue-sharing contract between brands, with an exogenous commission ratio λ, which is consistent with platform-based pricing research (Avinadav et al., 2022; Tsunoda & Zennyo, 2021). The two brands will reach the final alliance under different choices of knowledge management strategy, including the platform retail model based on brand knowledge integration, the interactive model of comprehensive knowledge management practices in the metaverse ecosystem, and a hybrid knowledge management strategy of the two models. The main parameter settings of our model are shown in Table 1.

In the retail model, the two brands only integrate their existing knowledge resources. The physical brand provides the specific product design details; the virtual brand offers platform services to the physical brand and launches virtual products identical with physical products. Consumers can receive the corresponding physical products or bundled products through the professional presentation of virtual products. The physical brand can expand sales through professional platform services such as professional presentation of product information, data support, and marketing promotion (Chen, Wang, & Wei, 2023; Wang et al., 2019). Especially virtual products can be contextualized or used for games to attract attention (Kshetri, 2022). At this point, the physical brand needs to pay different service fees to the virtual brand for different qualities of platform services. For simplicity, a quadratic function $s^2/2$ represents this service fee, where s denotes the investment of the physical brand for knowledge integration in adopting platform services (Alizadeh-Basban & Taleizadeh, 2020; Cai et al., 2022).



Fig. 1. Evolution of business models.

Table 1
Summary of notations.

Notation	Description
Parameters	
v	Consumer's basic perceived value for products
λ	Revenue-sharing ratio of virtual-physical alliance
θ	Fitness of products without interactive knowledge transmission
a	Level of technology investment in the metaverse ecosystem
k	Technology development cost of the metaverse ecosystem
s	Investment of the physical brand for knowledge integration
τ	Marginal utility brought by interactive experience knowledge transmission
γ	Marginal value that market knowledge contributed in the metaverse ecosystem
e	The service fee or subsidy for virtual-physical alliances in metaverse ecosystems
Variables	
D	Market demand
p	The price of the product
π_V^x	Profit of the virtual brand, where $x = R, IS, IT, HS, HT$
π_P^x	Profit of the physical brand, where $x = R, IS, IT, HS, HT$

In the metaverse ecosystem, the consumption scenario is no longer a simple physical transaction under the retail model but rather provides consumers with highly interactive purchasing experiences (Ahn et al., 2024; Chakraborty et al., 2024). Consumers can establish virtual identities containing personalized preference labels in virtual stores (Kshetri, 2022), use virtual avatars to try digital alternatives to traditional products, and obtain corresponding physical products. The acquisition and transmission of real-time data and knowledge provide continuous sales services for consumers and quickly respond to their personalized needs (Dolgui & Ivanov, 2022; Gu et al., 2021; Qi et al., 2023).

When the virtual brand intends to adopt the metaverse ecosystem to reach a strategic alliance with the physical brand, that is, a comprehensive knowledge management strategy in the interactive model, the virtual brand will charge a certain transfer payment e to the physical brand as a technology service fee. Simultaneously, the virtual brand developer needs to bear certain technology costs. Following Sun and Ji (2022), we assume that the virtual brand's investment level in metaverse technologies is a, so this part of the cost can be expressed as $ka^2/2$, where k denotes the technology development cost of the virtual brand. This quadratic cost function is widely used in technology investment (Shen et al., 2021). Besides the integration of existing knowledge between brands, we also consider the knowledge value created by the two-way brand-consumer interactions. Brands can transmit useful knowledge to consumers through metaverse technologies, such as delivering technology experiences that enhance consumers' utility and perceived value to maintain customer relationships (Chakraborty et al., 2024; Jafar et al., 2023; Pillai et al., 2023).

Moreover, brands can acquire market knowledge and data that can be processed to create additional sales value. In particular, as consumers interact with virtual avatars, brands can further sense changes in consumer preferences, offer customized products with security and privacy, and create personalized customer interactions (Gu et al., 2021; Modgil et al., 2022a, 2022b). We define the reverse value creation of acquiring market knowledge as the market knowledge contribution in the interactive metaverse and represent it as γa, where γ is the marginal value of the market knowledge contribution. The technology developer can directly benefit from this value. Therefore, to better achieve a strategic alliance with the physical brand, the virtual brand can decide to transform the service fee charged to the physical brand into a technology subsidy and strategically determine the charging strategy.

3.3. Market demand

Next, we will describe consumers' purchasing behaviors according to the consumer utility theory to determine game equilibrium decisions under the virtual and physical brands' market participations. We normalize the number of consumers entering the metaverse market into a unit, and when a brand offers two types of products or multiple channels simultaneously, consumers will strategically choose between them (Sun et al., 2022; Zhang et al., 2019). We also assume that if the brand provides both platform retail and virtual purchasing in the metaverse ecosystem, consumers will choose the one that maximizes channel utility. Each consumer can purchase the product of unit demand in one way at most. The valuation of online products provided by the brand for consumers is uncertain before trying on or experiencing them. Therefore, for the fitness of online products without interactive experience knowledge transmission $\theta(0 < \theta < 1)$, consumers can only obtain a rough valuation θv, where v represents the basic perceived value of the product by consumers and is evenly distributed on $[0, 1]$. The above assumptions have been widely applied in the relevant literature (Gao & Su, 2017; Song et al., 2022; Sun et al., 2022; Yang et al., 2021).

However, consumers can obtain the same purchasing experience as stores through metaverse technologies. They can create virtual identities and virtual shop assistant interactions in the metaverse ecosystem (Kshetri, 2022). Based on previous studies, firms can ensure transparency and consistency of information through digital showroom, physical showroom, and online and offline inventory (Li et al., 2020; Sun et al., 2022; Yang et al., 2020), and the metaverse interactive purchasing method also fully solves consumers' uncertainty about product information. Besides, previous studies suggested that consumers' willingness to continue purchasing is related to the purchasing experience brought by emerging technologies. The spatial environment characteristics they are in and their willingness to adopt new technologies play a key role; perceived value and utility and positive experiences affect their final purchasing decisions (Animesh et al., 2011; Chakraborty et al., 2024; Jafar et al., 2023; Pillai et al., 2023). In the metaverse ecosystem, consumers can enhance decision-making and obtain interactive technology experiences. Under this knowledge transmission, following Sun and Ji (2022) and Xin et al. (2024), we assume that the marginal utility brought by the transmission of technology experience knowledge is τ, so for the level of technology investment a, the total utility obtained by consumers can be expressed as τa.

3.4. Game timeline and parameter constraint

In the virtual-physical alliance game, the virtual brand with advanced technology is the dominant player, and the physical brand is the follower. The decision sequence of our game model is shown in Fig. 2. In stage 1, the virtual brand determines the ratio of the revenue-sharing contract, and the physical brand needs to share a certain proportion of the sales revenue; in stage 2, the virtual brand has the existing platform channels, and it also needs to choose whether to develop the metaverse ecosystem and whether to charge a service fee or provide a technology subsidy; in stage 3, the two brands reach a strategic alliance and choose the final knowledge management strategy; in stage 4, the virtual brand will determine the level of technology investment if the metaverse ecosystem is adopted; finally, the physical brand will decide on the platform service input if it exists and determines the price of the product.

To avoid some trivial situations and complex discussions, we summarized the relevant parameter constraints of the model, including the technology development cost, the revenue-sharing ratio set by the virtual brand, and service fees, to ensure that the model has a unique, non-negative optimal solution. The constraints of each parameter are discussed in detail in Appendix A.

4. Research methodology

4.1. Methodological overview

To explore the strategic dynamics of virtual-physical alliances

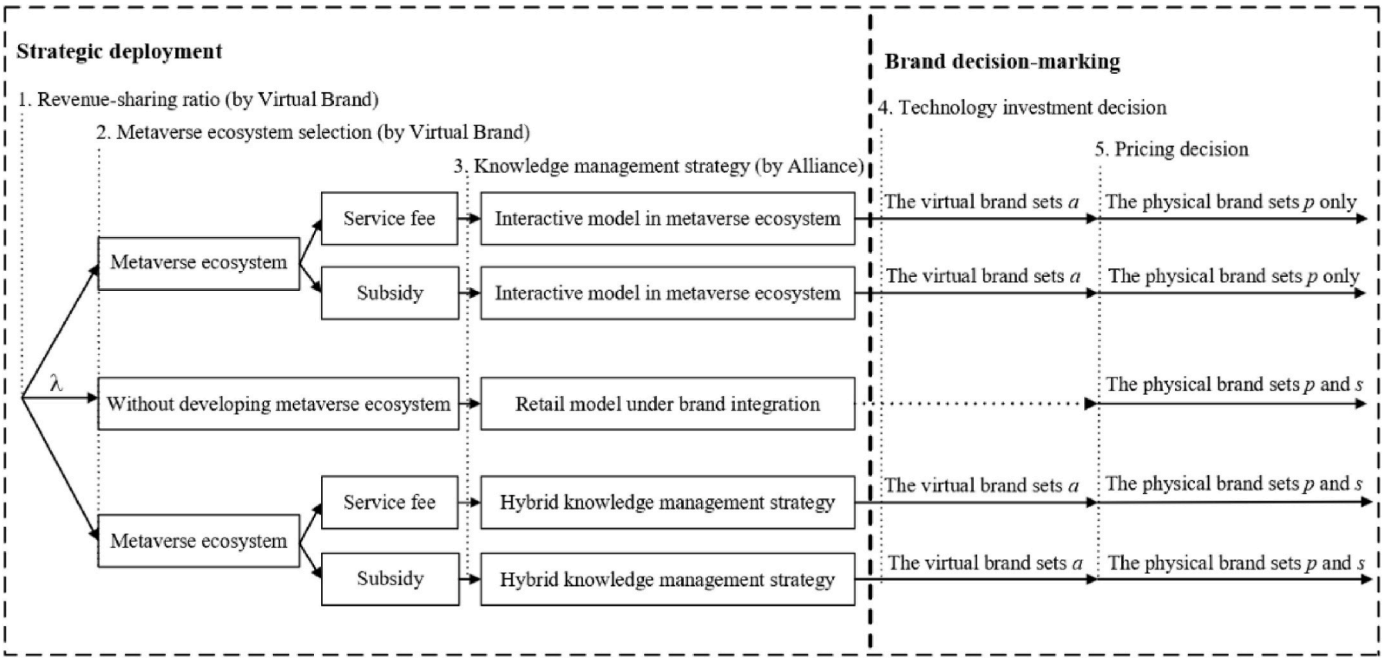


Fig. 2. Timeline of the game.

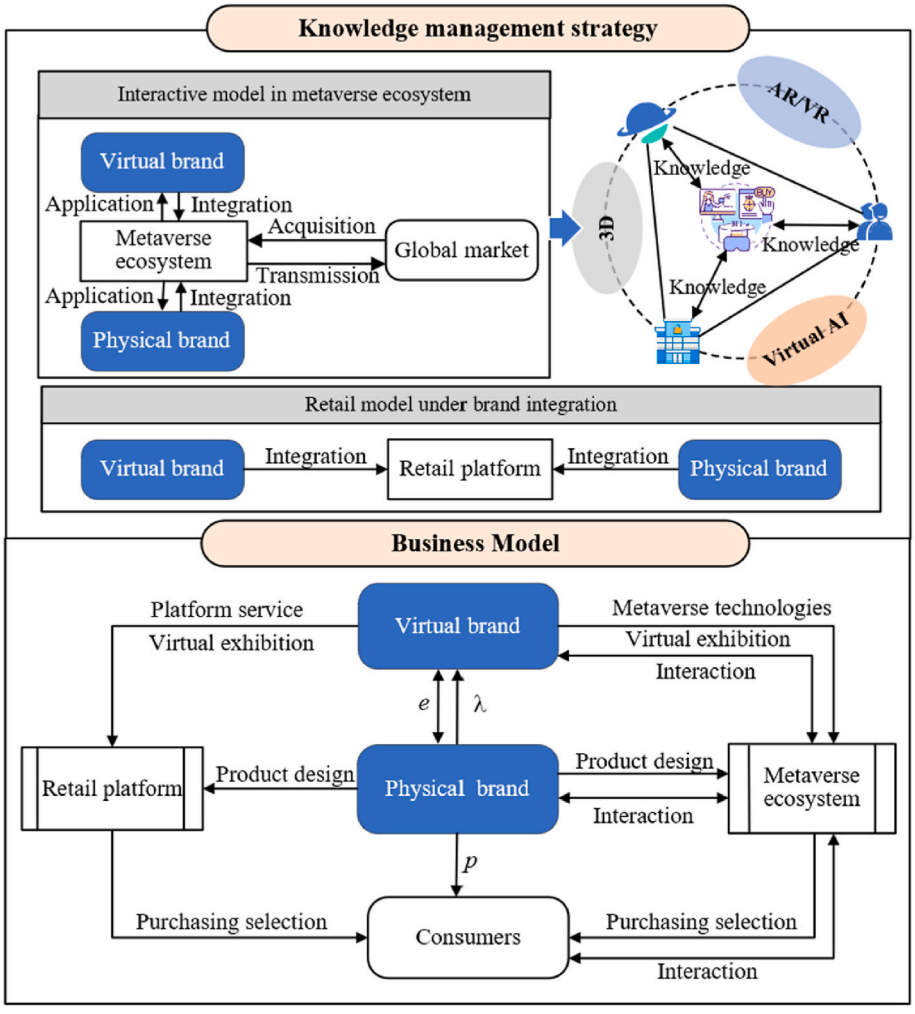


Fig. 3. The conceptual framework and model structure.

participating in knowledge management practices in the metaverse ecosystem and the deployment of optimal knowledge management strategies, we construct target revenue functions of the virtual and physical brands under different knowledge management strategies using game theory. The knowledge management strategies under each alliance mode include the retail mode under brand knowledge integration, the interactive mode in the metaverse ecosystem, and a hybrid knowledge management strategy of the two modes. The model structure for the above three scenarios is shown in Fig. 3. We use backward induction to solve game equilibriums, examine the virtual brand's and physical brand's strategic decisions, and ultimately determine the optimal knowledge management selection. For the presentation of the analysis results, we use the MATLAB software tool. First, we conduct sensitivity analyses on key parameters using numerical simulation to determine their impact on the result variables. Then, we solve for the feasible regions of each strategy based on parameter constraints to determine the optimal strategy selection. Finally, the key results of strategy selection are verified through a real case study.

4.2. Equilibrium analysis and numerical simulation

4.2.1. Retail model under brand integration

In Strategy R, the physical brand and virtual brand only integrate existing brand knowledge resources. The physical brand provides content and details of product design, and the leading virtual brand offers online channels containing market resources. With this brand alliance, they can provide better platform services for consumers through professional virtual product exhibitions. Because in the retail mode, consumers cannot perceive the fitness of products completely, the consumer's product value evaluation can be expressed as θv. In the process of brand integration, the higher the investment in virtual brand platform services s, the higher the utility of consumer product purchases. At this point, the net surplus of consumers can be expressed as $\theta v - p + s$, and the total market demand is $1-(p-s)/\theta$. When the brand alliance chooses the retail model under brand knowledge integration, the physical brand needs to pay a service fee to the virtual brand as an investment in platform services, i.e., $s^2/2$. Then, the profit functions of the virtual brand and physical brand are summarized as follows:

$$\pi_V^R = \lambda p\left(1-\frac{p-s}{\theta}\right)+\frac{1}{2}s^2 \text{ and } \pi_P^R = (1-\lambda)p\left(1-\frac{p-s}{\theta}\right)-\frac{1}{2}s^2. \tag{1}$$

Proposition 1. *In Strategy R, the optimal equilibriums of the product price p^{R^*}, the investment of the physical brand for knowledge integration s^{R^*}, and the profits of the virtual brand and physical brand $\pi_V^{R^*}$ and $\pi_P^{R^*}$ are shown in (2)-(5), respectively.*

$$p^{R^*} = \frac{\theta^2}{2\theta+\lambda-1}, \tag{2}$$

$$s^{R^*} = \frac{\theta-\theta\lambda}{2\theta+\lambda-1}, \tag{3}$$

$$\pi_V^{R^*} = \frac{\theta^2(1+\lambda(2\theta+\lambda-2))}{2(2\theta+\lambda-1)^2}, \tag{4}$$

$$\pi_P^{R^*} = \frac{\theta^2(1-\lambda)}{2(2\theta+\lambda-1)}. \tag{5}$$

Accordingly, some findings we obtain by checking the first-order derivative of the above optimal solutions are shown in Lemma 1. We also use two numerical examples to depict the results, as shown in Fig. 4.

Lemma 1. *Under Strategy R, we have*:

(i) *The product price p^{R^*} will increase with the product fitness θ if and only if $\theta > 1-\lambda$, i.e., $\partial p^{R^*}/\partial\theta > 0$;*
(ii) *The investment of the physical brand for knowledge integration s^{R^*} will decrease with θ, i.e., $\partial s^{R^*}/\partial\theta < 0$;*
(iii) *The profits of the virtual brand and physical brand $\pi_V^{R^*}$ and $\pi_P^{R^*}$ vary non-monotonically with parameter θ, and $\pi_P^{R^*}$ will also increase with θ if and only if $\theta > 1-\lambda$.*

Lemma 1 indicates that the product price first decreases and then increases with the increase of product fitness. Because the cost of knowledge integration s, the physical brand will decide on a higher product price in the initial stage. Consumers will face more uncertainty if the fitness of products is lower than the general level. Only consumers with a low valuation of the product or low willingness to pay are willing to buy the products. This moment, the physical brand will decrease the price to expand demand and attract more consumers. When the product fitness is high, the physical brand will increase the product price to obtain more revenues in the face of consumers' high willingness to pay, which is also illustrated by the increase in the physical brand's profit, as shown in Lemma 1(iii).

Lemma 1(ii) shows that the investment of the physical brand for the knowledge integration, that is, the investment for the virtual brand's platform services, will decrease as the product fitness increases. The reason is that investment for platform services can also improve consumers' willingness to pay. However, as a service input, it also represents a cost loss so a high investment level for knowledge integration will hurt the physical brand's total profit. Therefore, as long as the level of product fitness increases product demand, the physical brand can

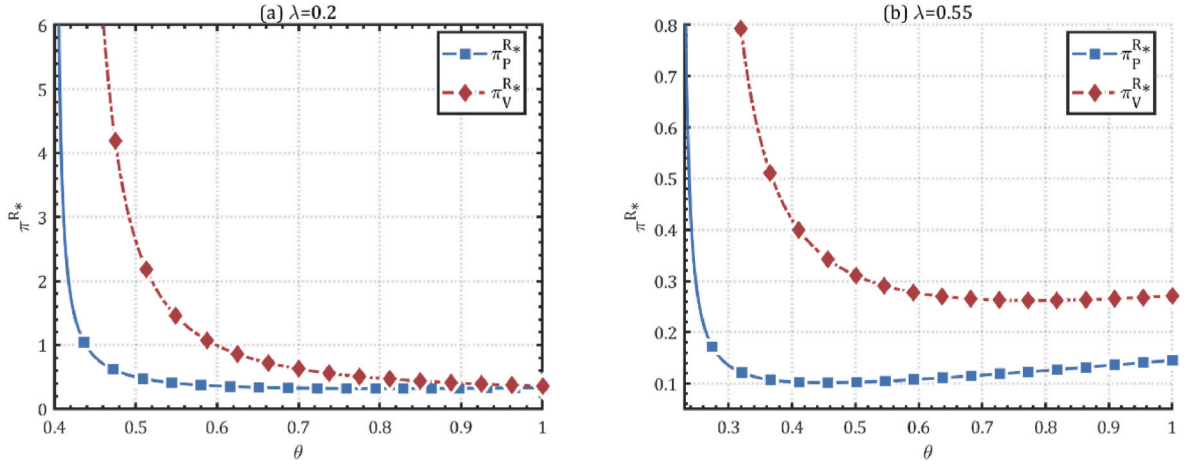


Fig. 4. The non-monotonic trends of the virtual brand's and the physical brand's profits.

appropriately reduce the service input. Similarly, the change in product price and total consumer demand also means that the virtual brand's profit varies non-monotonically. We also use two numerical examples with settings $\lambda = 0.2$ and $\lambda = 0.55$. Fig. 4 shows the non-monotonic trends of the virtual brand's and the physical brand's profits $\pi_V^{R^*}$ and $\pi_P^{R^*}$.

Fig. 4 depicts that the profits under the lower revenue-sharing ratio vary more with parameter θ than the profits under the higher revenue-sharing ratio, and the virtual brand's profit is higher under the same product fitness level. And if and only if the revenue-sharing ratio set by the virtual brand is high, the virtual brand's profit will increase with parameter θ; otherwise, the virtual brand's profit will continually decrease. The reason is that when the revenue-sharing ratio set by the virtual brand is low, the physical brand who is more favorable in the revenue-sharing contract will significantly increase the product price to retain consumers who are more willing to pay and obtain more revenue. And although consumer demand will rise as the parameter θ increases, it will not be enough to offset the loss of profit for the weaker party. Therefore, only a higher revenue-sharing ratio level is beneficial to the virtual brand. Furthermore, Fig. 4 also shows that in Strategy R, regardless of the revenue-sharing ratio, the profit of the virtual brand is always higher than that of the physical brand, i.e., $\pi_V^{R^*} > \pi_P^{R^*}$, which suggests that the virtual brand can benefit more from the retail model.

4.2.2. Interactive model in metaverse ecosystem

When the brand alliance chooses the interactive model, the two brands will integrate the existing knowledge resources, and the virtual brand also intends to adopt metaverse technologies to create an interactive metaverse ecosystem space to sell products. Consumers can create virtual identities and interact with virtual avatars to virtually experience and try on products. Through various metaverse technologies, the brand can acquire and apply market knowledge, especially personalized needs and preferences. Simultaneously, the interactive experience brought by technologies can also be transmitted to consumers. Then, they can create more sustainable performance under a comprehensive knowledge management model. In an interactive model, the physical brand also needs to share a certain proportion of revenue with the virtual brand and pay a specific service fee or receive a technology subsidy offered by the virtual brand. We will explore the related decisions of the virtual and physical brands in both cases with different charging strategies.

Interactive model with a service fee. In Strategy IS, the virtual brand provides the metaverse ecosystem for their brand alliance, and consumers can set up their virtual avatars in the virtual store to experience more suitable products. Therefore, the unfitness of the product can be eliminated, i.e., the product fitness is $\theta = 1$. Additionally, consumers can obtain an interactive technology experience, which is related to the level of technology investment the virtual brand determines. We can express this part of consumer utility brought by experience knowledge transmission as τa. Then the consumer's total surplus of the product can be expressed as $v + \tau a - p$, and the consumer demand is $D = 1 - p + \tau a$. In this strategy, the virtual brand can obtain the service fee e and the market knowledge contributed value, i.e., γaD. Simultaneously, the virtual brand also needs to pay a particular technology cost $ka^2/2$ as an input in metaverse technologies. Under the revenue-sharing contract, the profit functions of the physical brand and the virtual brand can be expressed as:

$$\pi_V^{IS} = (\lambda p + \gamma a + e)(1 - p + \tau a) - \frac{1}{2}ka^2 \text{ and } \pi_P^{IS} = ((1-\lambda)p - e)(1 - p + \tau a). \tag{6}$$

Proposition 2. *When the physical brand must pay a service fee for the alliance, the optimal equilibriums of the product price p^{IS^*}, the level of technology investment a^{IS^*}, and the profits of the virtual brand and the physical brand $\pi_V^{IS^*}$ and $\pi_P^{IS^*}$ are shown in (7)-(10), respectively.*

$$p^{IS^*} = \frac{2k(1+e-\lambda) + \tau(\gamma(\lambda - 1 - 3e) + e(1-2\lambda)\tau)}{2(1-\lambda)(2k - \tau(2\gamma + \lambda\tau))}, \tag{7}$$

$$a^{IS^*} = \frac{\gamma(e + \lambda - 1) + (\lambda - 1)(e + \lambda)\tau}{(1-\lambda)(\tau(2\gamma + \lambda\tau) - 2k)}, \tag{8}$$

$$\pi_V^{IS^*} = \frac{\left[\begin{array}{l}(1-\lambda)^2(\gamma^2 + 2k\lambda) + e^2(2k(\lambda - 2) + (\gamma + \tau)^2) \\ +2e(\lambda - 1)(2k(\lambda - 1) + \gamma(\gamma + \tau))\end{array}\right]}{4(1-\lambda)^2(2k - \tau(2\gamma + \lambda\tau))}, \tag{9}$$

$$\pi_P^{IS^*} = \frac{(\tau(\gamma(-1 + e + \lambda) + e\tau) - 2k(-1 + e + \lambda))^2}{4(1-\lambda)(\tau(2\gamma + \lambda\tau) - 2k)^2}. \tag{10}$$

Interactive model with a technology subsidy. In Strategy IT, the total market demand is still $D = 1 - p + \tau a$. If the metaverse ecosystem is adopted for brand alliance, the physical brand can obtain the technology subsidy e offered by the virtual brand as a contribution to reach the virtual-physical alliance. Accordingly, the profit functions of the virtual brand and the physical brand can be summarized as follows:

$$\pi_V^{IT} = (\lambda p + \gamma a - e)(1 - p + \tau a) - \frac{1}{2}ka^2 \text{ and } \pi_P^{IT} = ((1-\lambda)p + e)(1 - p + \tau a). \tag{11}$$

Proposition 3. *When the physical brand can get a technology subsidy, the optimal equilibriums of the product price p^{IT^*}, the level of technology investment a^{IT^*}, and the profits of the virtual brand and the physical brand $\pi_V^{IT^*}$ and $\pi_P^{IT^*}$ are shown in (12)-(15), respectively.*

$$p^{IT^*} = \frac{\tau(\gamma(3e + \lambda - 1) + e(2\lambda - 1)\tau) - 2k(e + \lambda - 1)}{2(1-\lambda)(2k - \tau(2\gamma + \lambda\tau))}, \tag{12}$$

$$a^{IT^*} = \frac{\gamma(1 + e - \lambda) + (e - \lambda)(\lambda - 1)\tau}{(1-\lambda)(2k - \tau(2\gamma + \lambda\tau))}, \tag{13}$$

$$\pi_V^{IT^*} = \frac{\left[\begin{array}{l}(1-\lambda)^2(\gamma^2 + 2k\lambda) + e^2(2k(\lambda - 2) + (\gamma + \tau)^2) \\ +2e(1-\lambda)(2k(\lambda - 1) + \gamma(\gamma + \tau))\end{array}\right]}{4(1-\lambda)^2(2k - \tau(2\gamma + \lambda\tau))}, \tag{14}$$

$$\pi_P^{IT^*} = \frac{(-2k(1 + e - \lambda) + \tau(\gamma(1 + e - \lambda) + e\tau))^2}{4(1-\lambda)(\tau(2\gamma + \lambda\tau) - 2k)^2}. \tag{15}$$

Mathematically, we also derive some findings based on the above optimal outcomes in adopting an interactive model, and the analytical results are shown in Lemma 2.

Lemma 2. *The effects of related parameters on the equilibriums are stated in Table 2*:

Lemma 2 depicts the value created by the adoption of an interactive model for the brand alliance. Specifically, as the marginal utility brought by interactive experience knowledge transmission τ increases, and the marginal value that market knowledge contributed γ increases, as well as the development cost of metaverse technologies decreases, the price of the product, the decision level of technology investment, and the profits of the virtual brand and the physical brand all increase. In

Table 2
The value created by the adoption of an interactive model for the brand alliance.

	p^{IS^*}	p^{IT^*}	a^{IS^*}	a^{IT^*}	$\pi_V^{IS^*}$	$\pi_V^{IT^*}$	$\pi_P^{IS^*}$	$\pi_P^{IT^*}$
τ	↑	↑	↑	↑	↑	↑	↑	↑
γ	↑	↑	↑	↑	↑	↑	↑	↑
e	↑	↓	↕	↕	↕	↕	↕	↕
k	↓	↓	↓	↓	↓	↓	↓	↓

Note: ↑ *indicates increasing*; ↓ *indicates decreasing*; ↕ *indicates non-monotonic*.

comprehensive knowledge management contribution, a higher parameter τ means that consumers can get a better technology experience from knowledge transmissions, increasing their willingness to pay and consumer demand. The physical brand has more confidence to raise the product price to obtain more revenues, and the virtual brand can also get more through the revenue sharing contract and correspondingly increase their investment in metaverse technologies to improve the decision level of technology investment. Then, as the decision level improves, it will further affect the physical brand's pricing decisions.

Due to the existence of contribution value form market knowledge, the virtual brand will decide whether to provide technology subsidies or collect service fees from the physical brand. Therefore, the profits of the physical brand and the virtual brand will be affected by multiple factors and exhibit non-monotonic variations. When the values created by the metaverse ecosystem are insufficient to offset the losses caused by the delivery of fees and costs, their profits will decrease. At this point, the virtual brand will strategically adjust the decision level of technology investment. For the pricing decisions, when the physical brand obtains the technology subsidy from the virtual brand, it is equivalent to indirectly giving a subsidy to consumers due to market knowledge acquisitions, and the physical brand has the incentive to reduce the price to expand the demand. However, when it is necessary to pay the service fee, the physical brand will face a profit loss. The physical brand will raise the price to retain consumers with a higher willingness to pay, thus covering its costs.

Because it is difficult to specify the change in the level of technology investment, we still use numerical examples to illustrate the results of the above non-monotonic changes and set the related parameters to: $\lambda = 0.2$ and (*a*) I: $\tau > \gamma$ ($\tau = 3.8$; $k = 18.6$); II: $\tau < \gamma$ ($\tau = 2.5$, $k = 12$), and (*b*) I: $\tau > \gamma$ ($\tau = 1.3$; $k = 2.12$); II: $\tau < \gamma$ ($\tau = 0.5$, $k = 1$), respectively.

Based on Fig. 5, under different technology experience utility τ and technology development cost k, the effects of transfer payment fee e on the level of investment decisions are opposite. When parameter τ is high, if the virtual brand charges the physical brand a service fee, then with the fee increasing, the virtual brand will correspondingly improve the decision level of technology investment. However, when the virtual brand pays a corresponding subsidy as a contribution, as parameter e increases, it will reduce the investment level in the decision process. The virtual brand can strategically adjust the level of technology investment, first to avoid more cost losses, i.e., $ka^2/2$, and second to obtain the market knowledge contribution, i.e., γaD.

When parameter τ is high, although it can play a positive role in technology experience utility, it will incur higher technology development cost $k = 18.6$, and the virtual brand will maintain a low level of technology investment. In Strategy IS, the virtual brand can obtain a portion of service fees. As the fee increases, the virtual brand will slightly improve its decision-making level, while Strategy IT has the opposite performance. When the value of the parameter τ is low, the development cost is also low $k = 12$, so the virtual brand will have the confidence to decide on a high level of technology investment with a low-profit loss. As the physical brand faces the service fee, it will increase the product price to cover the cost, reducing total market demand. If the virtual brand does not take measures now, the higher level of investment will continue to increase the product price (i.e., $p^{AM^*} = (1 + e/1 - \lambda + \tau a^{AM^*})/2$). The demand will be further reduced, so as e increases, the virtual brand will reduce the level of technology investment to offset a reduction in profit due to the low demand. The technology subsidy from the virtual brand indirectly subsidizes consumers for contributing market knowledge, so the product price will remain relatively low, thus expanding total demand. The virtual brand will further improve the level of technology investment to increase knowledge contribution value.

By comparing the equilibrium results under these two strategies, we also derive Corollary 1 as follows.

Corollary 1. *The equilibrium outcomes in Strategies IS and IT satisfy*:

(i) *If the parameters satisfy that* $\tau > \gamma$ *and* $\lambda < (\tau - \gamma)/\tau$, *the level of technology investment of Strategy IS will be higher than that of Strategy IT, i.e.,* $a^{IS^*} > a^{IT^*}$.

(ii) *The product price of Strategy IS is higher than that of Strategy IT, i.e.,* $p^{IS^*} > p^{IT^*}$.

(iii) *If the parameters satisfy that* $\tau > 2\gamma$, $\lambda < (\tau - 2\gamma)/\tau$ *and* $k < (\gamma\tau + \tau^2)/2$, *the physical brand's profit and consumer demand of Strategy IS will be higher than those of Strategy IT, i.e.,* $\pi_P^{IS^*} > \pi_P^{IT^*}$ *and* $D^{IS^*} > D^{IT^*}$.

(iv) *If the parameters satisfy that* $\lambda > \lambda_p$, $k < \gamma(\gamma + \tau)/(2 - 2\lambda)$ *and* $\tau \geq \gamma/2$ *or* $\tau < \gamma/2$ *and* $k < \gamma(\gamma + \tau)/(2 - 2\lambda)$, *the virtual brand's profit of Strategy IT will be higher than that of Strategy IS, i.e.,* $\pi_V^{IT^*} > \pi_V^{IS^*}$.

The expression of λ_p is defined in Appendix B.3.

Corollary 1 (i) shows that when the marginal utility from interactive experience is high, and the revenue-sharing ratio is low, the virtual brand's decision level when offering a subsidy is lower than that when receiving a service fee. Consistent with the analysis in Lemma 2, the level of technology investment will lead to opposite decisions in different situations due to varying technology experience utility. Besides, the two charging situations also determine the opposite variations in the level of technology investment (as presented in Fig. 5). The higher price in strategy IS reduces the product demand, and if the virtual brand still decides on a higher level of technology investment, the demand will be further reduced. In Strategy IT, the physical brand will determine a lower product price to expand demand, and the virtual brand can also

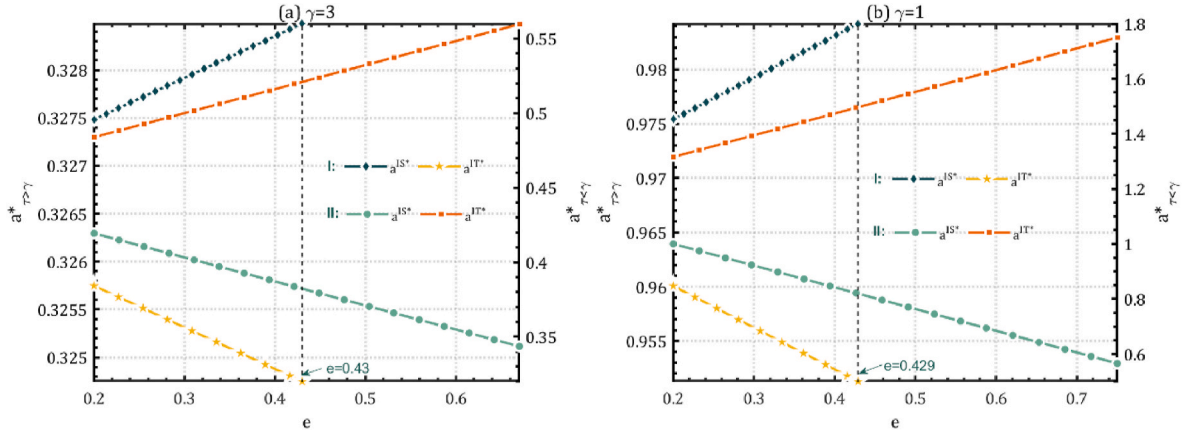


Fig. 5. The impact of transfer payment fee e on the technology investment level.

give full play to the market knowledge contribution by determining a higher level of technology investment, as shown in Lemma 2. It is also consistent with the price comparison in Corollary 1 (ii). When consumers' technology utility perceived by knowledge transmission is high, the reduction in demand will be alleviated to a certain extent. Then the decision level in Strategy IS will be higher than in Strategy IT, and when the revenue-sharing ratio is low, the same is true. A lower revenue-sharing ratio means that the physical brand will retain more revenue, which is unfavorable to the virtual brand, and the virtual brand also has to pay a technology subsidy for the physical brand. The technology utility is higher, and the virtual brand will face higher technology development costs. In the case of multi-part profit losses, the investment level determined by the virtual brand is lower than that in Strategy IS.

Corollaries 2 (iii) and (iv) show the impact of related parameters on consumer demand and the virtual brand's and the physical brand's profits in both strategies. If and only if consumers' technology utility is high, and the revenue-sharing ratio and technology development cost is low, the consumer demand and the profit of the physical brand in Strategy IT are lower than those in Strategy IS. Because the technology subsidy indirectly subsidizes consumers in Strategy IT, the product price in Strategy IS is always higher than that in Strategy IT. In most cases, the total consumer demand in Strategy IS will be lower than that in Strategy IT. Besides the impact of the product price, the level of technology investment and technology utility will further affect total demand. As depicted in Corollary 1 (i), the optimal decision in Strategy IT is lower than that in Strategy IS only when the consumer's utility perceived by knowledge transmission is high, and the revenue-sharing ratio is low. Furthermore, choosing Strategy IS is unfavorable when the technology development cost is low, as the virtual brand may still determine a higher investment level, resulting in further reductions in consumer demand. In summary, it is favorable for both the virtual brand and the physical brand to choose Strategy IT. Corollary 1 (iv) also indicates that, in most cases, the virtual brand's profit under Strategy IT will be higher than that of Strategy IS.

Fig. 6 also shows the non-monotonic impact of parameter e on the profits of the virtual brand and the physical brand and depicts the comparisons between the high and low-profit levels of the two charging scenarios. We set the related parameters to: $\lambda = 0.2$, $\gamma = 3$, and I: $\tau = 1.3$, $k = 6.1$, and II: $\tau = 1.3$, $k = 7.5$, and III: $\tau = 3.8$, $k = 18.6$, and IV: $\tau = 2.5$, $k = 12$, and V: $\tau = 7.6$, $k = 40$, and VI: $\tau = 0.8$ and $k = 3.7$.

Fig. 6 shows that with the increase of parameter e, the profits of the virtual brand and the physical brand will vary non-monotonically under different scenarios. Specifically, Fig. 6(b) and (d) indicate that the profit of the physical brand under Strategy IS decreases in most cases as the service fee increases. The physical brand has decided to cover costs by increasing the product price. When consumer's utility perceived by technology experience transmission is higher, the physical brand's profit may show a downward trend, because it is difficult for the physical brand to offset the loss of profit at this time. The profit will increase if the higher utility positively affects the demand to offset the loss caused by a

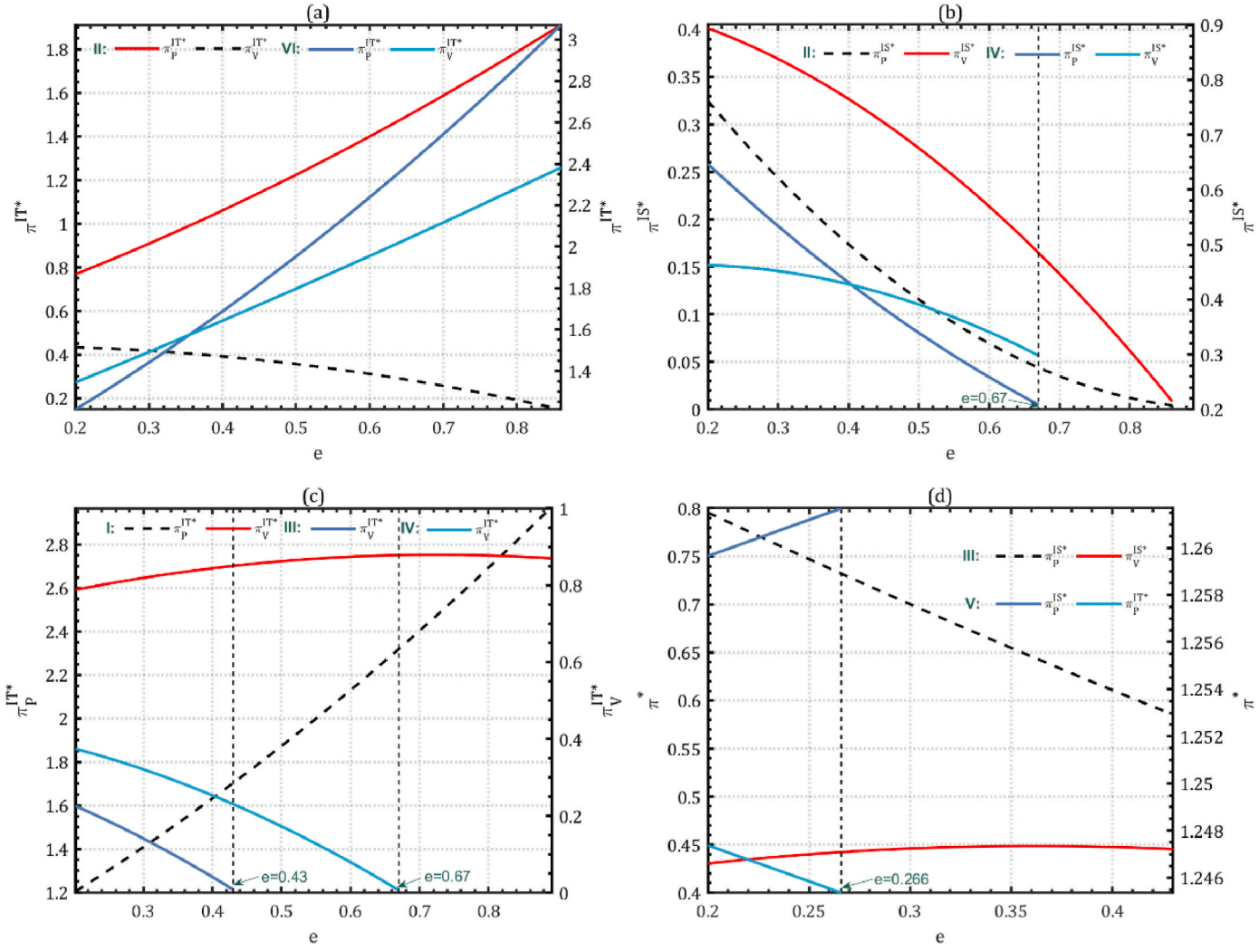


Fig. 6. The impact of transfer payment fees on the profits of the virtual brand and the physical brand.

higher technology development cost. From Fig. 6(b) and (d), we find that the virtual brand's profit in Strategy IS also varies non-monotonically. In most cases, the profit decreases with the service fee. Still, when consumers' technology utility is higher, i.e., $\tau > \gamma$, the virtual brand's profit will show a trend of increasing and then decreasing. The intuitive reason is that a higher product price in Strategy IS leads to lower demand, and the physical brand can cover profit losses through this decision under certain circumstances. However, the virtual brand still faces a technology development cost, and the revenue-sharing contract cannot provide a higher revenue share, resulting in lower profit. When consumers' technology utility is higher, the technology development cost is also higher, and the virtual brand will decide on a lower level of technology investment, as presented in Fig. 5, to ensure a lower price, thereby alleviating the demand reduction. However, the price may still be higher than the average level, and as the losses caused by the fee and other costs increase, the virtual brand's profit still presents a downward trend.

Similarly, Fig. 6(a)–(c) and (d) demonstrate that in most cases, when the parameter τ is low, the physical brand's profit in Strategy IT will increase with the technology subsidy. However, the physical brand's profit will gradually decrease when technology costs are too high. In Strategy IT, the technology subsidy can be used as an indirect subsidy to consumers contributing knowledge acquisition and application, so the price decrease expands the demand, and the physical brand's profit increases in most cases. The variation in the decline indicates that the higher technology cost indirectly leads to a decrease in the physical brand's profit, which is consistent with the previous analysis. In Strategy IT, Fig. 6(a) and (c) show that under the numerical systems II, VI, and I, the virtual brand's profit will correspondingly show a downward trend, an upward trend, and a first upward trend followed by a downward trend. They intuitively show that with low technology utility, the development cost will significantly affect the virtual brand's profit. With the technology cost increases, the virtual brand's profit gradually shows a trend from rising then falling. Moreover, Fig. 6(c) also shows that although consumers' technology utility is higher, the virtual brand's profit still decreases with the increase in the technology subsidy due to the impact of the high technology cost.

Furthermore, in Strategy IS, Fig. 6(b) and (d) show that a lower technology development cost is more favorable for the virtual brand, which can obtain more revenues than the physical brand. However, with the increase in the consumer's technology utility, the technology development cost almost doubles, resulting in the virtual brand's profit being lower than that of the physical brand. Still, as the service fee increases, the physical brand's profit is lower than the virtual brand's profit due to its more significant reduction. While with the rise in technology utility and the technology cost, that is, the cost is reached at $k = 18.6$, the virtual brand's profit will always be lower than that of the physical brand. When the alliance chooses the technology subsidy, it can be found from Fig. 6(a) and (c) that in most cases, the physical brand's profit is higher than that of the virtual brand. If and only if the technology utility and technology development cost are low, that is, the cost is only at $k = 3.7$, then the virtual brand's profit is higher than that of the physical brand. With the subsidy increases, the physical brand's performance increase is more considerable, and its profit is higher than that of the virtual brand. Therefore, if the virtual brand pays the technology subsidy, the physical brand can benefit more than the virtual brand. And as stated in Corollary 1, their profits will all be higher under Strategy IT, which is also proved in Fig. 6(a).

4.2.3. Hybrid knowledge management strategy

In the hybrid strategy, the virtual-physical alliance will simultaneously provide two models, i.e., the interactive model in the metaverse ecosystem and the retail model under brand integration. Consumers can make a utility-maximizing decision between two purchasing selections. When they experience the retail model, they still face the product unfitness, so their valuation is θv. On the contrary, consumers entering the metaverse ecosystem can gain interactive technology experiences and choose the most suitable products, so the perceived product value for these consumers is v. According to the input of physical brand and virtual brand in knowledge integration and management strategy, that is, the investment levels of platform services and metaverse technologies, consumers can obtain the value s and τa, respectively. Then their surplus can be expressed as $\theta v + s - p$ and $v + \tau a - p$, respectively. Therefore, the market demand under the hybrid strategy is as follows:

$$\begin{cases} D_R = \int_{v:\theta v+s-p>\max\{0,v+\tau a-p\}} f(v)dv, \\ D_I = \int_{v:v+\tau a-p>\max\{0,\theta v+s-p\}} f(v)dv. \end{cases} \tag{16}$$

Hybrid strategy with a service fee. In Strategy HS, the physical brand needs to pay service fees to strategic alliances with the leading virtual brand and decide the product price and the investment level of platform services. The virtual brand needs to decide the level of technology investment and pay for the development costs of metaverse ecosystem. Under the revenue-sharing contract, the profit functions of the virtual brand and the physical brand can be formulated as follows:

$$\begin{aligned} \pi_V^{HS} &= \lambda p D_R + (\lambda p + \gamma a + e) D_I + \frac{1}{2}s^2 - \frac{1}{2}ka^2 \text{ and } \pi_P^{HS} \\ &= (1-\lambda) p D_R + ((1-\lambda)p - e) D_I - \frac{1}{2}s^2. \end{aligned} \tag{17}$$

Proposition 4. *When the physical brand needs to pay a service fee in the hybrid strategy, the optimal equilibriums of the product price p^{HS*}, the investment levels of platform services and metaverse technologies s^{HS*}, a^{HS*} and the profits of the virtual brand and the physical brand π_V^{HS*} and π_P^{HS*} are shown in (18)-(22), respectively.*

$$p^{HS*} = \frac{\theta(e + \theta - \theta^2)}{(1-\theta)(2\theta + \lambda - 1)}, \tag{18}$$

$$s^{HS*} = \frac{\theta(1 + 2e - \theta - \lambda + \theta\lambda)}{(1-\theta)(2\theta + \lambda - 1)}, \tag{19}$$

$$a^{HS*} = \frac{\gamma(\theta(3 - 2e + 2(\theta-2)\theta - \lambda) + \lambda - 1) + e(1-\theta)(2\theta + \lambda - 1)\tau}{(1-\theta)(2\theta + \lambda - 1)(k(1-\theta) - 2\gamma\tau)}, \tag{20}$$

$$\pi_V^{HS*} = \frac{\begin{bmatrix} (1-\theta)^2\left(2\gamma(\theta-1)\theta^2(1+\lambda(2\theta+\lambda-2))\tau + \gamma^2(2(1-\theta)\theta + \lambda - 1)^2 + k(1-\theta)^2\theta^2(1+\lambda(2\theta+\lambda-2))\right) \\ +2e\gamma(1-\theta)\left(2\gamma\theta(1+2(\theta-1)\theta-\lambda) + (1-\theta)\left(4\theta^3 + 2\theta^2(\lambda-5) + 4\theta(1-\lambda) - (1-\lambda)^2\right)\tau\right) \\ +e^2\left(4\gamma^2\theta^2 - 2k(1-\theta)^2\theta(2\theta+\lambda-2) + 4\gamma(\theta-1)\theta\tau + (1-\theta)^2(2\theta+\lambda-1)^2\tau^2\right) \\ -2ke(1-\theta)^3\left(4(1-\theta)^2\theta + 2\lambda + 2(\theta-2)\theta\lambda - \lambda^2 - 1\right) \end{bmatrix}}{2(1-\theta)^3(2\theta+\lambda-1)^2(k(1-\theta) - 2\gamma\tau)}, \tag{21}$$

$$\pi_P^{HS^*}=\frac{\left[\begin{array}{l}(\theta-1)\theta^2(\lambda-1)(k(\theta-1)+2\gamma\tau)+2e^2\left((2\theta+\lambda-1)\tau^2-k\theta\right)\\+2e(1+2(\theta-1)\theta-\lambda)(k(\theta-1)+\gamma\tau)\end{array}\right]}{2(1-\theta)(2\theta+\lambda-1)(k(\theta-1)+2\gamma\tau)}. \tag{22}$$

Hybrid strategy with a technology subsidy. In Strategy HT, the virtual brand pays the technology subsidy to the physical brand, and the physical brand only needs to pay the input of platform services. Other decisions of the physical brand and the virtual brand are consistent with those in Strategy HS, so their profit functions can be expressed as follows:

$$\begin{aligned}\pi_V^{HT}&=\lambda pD_R+(\lambda p+\gamma a-e)D_I+\frac{1}{2}s^2-\frac{1}{2}ka^2 \text{ and } \pi_P^{HT}\\&=(1-\lambda)pD_R+((1-\lambda)p+e)D_I-\frac{1}{2}s^2.\end{aligned} \tag{23}$$

Proposition 5. *When the physical brand gets a technology subsidy for the alliance in the hybrid strategy, the optimal equilibriums of the product price p^{HT^*}, the investment levels of platform services and metaverse technologies s^{HT^*}, a^{HT^*} and the profits of the virtual brand and the physical brand $\pi_V^{HT^*}$ and $\pi_P^{HT^*}$ are shown in (24)-(28), respectively.*

$$p^{HT^*}=\frac{\theta(e+(\theta-1)\theta)}{(\theta-1)(2\theta+\lambda-1)}, \tag{24}$$

$$s^{HT^*}=\frac{\theta(2e+\theta+\lambda-\theta\lambda-1)}{(\theta-1)(2\theta+\lambda-1)}, \tag{25}$$

$$a^{HT^*}=\frac{\gamma(\theta(3+2e+2(\theta-2)\theta-\lambda)+\lambda-1)+e(\theta-1)(2\theta+\lambda-1)\tau}{(\theta-1)(2\theta+\lambda-1)(k(\theta-1)+2\gamma\tau)}, \tag{26}$$

$$\pi_V^{HT^*}=\frac{\left[\begin{array}{l}(1-\theta)^2\left(\gamma^2(2(1-\theta)\theta+\lambda-1)^2+k(1-\theta)^2\theta^2(1+\lambda(2\theta+\lambda-2))+2\gamma(\theta-1)\theta^2(1+\lambda(2\theta+\lambda-2))\tau\right)\\+2e\gamma(1-\theta)\left(2\gamma\theta(2(1-\theta)\theta+\lambda-1)+(\theta-1)\left(4\theta^3+2\theta^2(\lambda-5)+4\theta(1-\lambda)-(1-\lambda)^2\right)\tau\right)\\+e^2\left(4\gamma^2\theta^2-2k(1-\theta)^2\theta(2\theta+\lambda-2)+4\gamma(\theta-1)\theta\tau+(1-\theta)^2(2\theta+\lambda-1)^2\tau^2\right)\\+2ek(1-\theta)^3\left(4(1-\theta)^2\theta+2\lambda+2(\theta-2)\theta\lambda-\lambda^2-1\right)\end{array}\right]}{2(\theta-1)^3(2\theta+\lambda-1)^2(k(\theta-1)+2\gamma\tau)}, \tag{27}$$

$$\pi_P^{HT^*}=\frac{\left[\begin{array}{l}(1-\theta)\theta^2(\lambda-1)(k(\theta-1)+2\gamma\tau)+2e^2\left(k\theta-(2\theta+\lambda-1)\tau^2\right)\\+2e(1+2(\theta-1)\theta-\lambda)(k(\theta-1)+\gamma\tau)\end{array}\right]}{2(\theta-1)(2\theta+\lambda-1)(k(\theta-1)+2\gamma\tau)}. \tag{28}$$

Similarly, we can derive Lemma 3 by checking the first-order partial derivative of the equilibrium results in the hybrid strategies.

Lemma 3. *The effects of related parameters on the equilibriums are stated in* Table 3.

Lemma 3 demonstrates the robustness of our previous parametric analysis. As knowledge contribution increases brought by comprehensive knowledge management in an interactive model, the level of technology investment and the profits of the virtual brand and the physical brand will increase. That is the rise in consumers' technology utility perceived by experience knowledge transmission and marginal value brought by market knowledge acquisition and application as well as the decrease in the technology development cost for strategic alliance, as presented in Lemma 2. The physical brand's decisions are related to the total demand to buy products, and the consumer demand at this time is numerically equal to the market when all consumers choose the retail model (i.e., $1-(p-s)/\theta$). Consumers' willingness to enter the metaverse ecosystem depends on the level of technology investment, and the physical brand's decisions have nothing to do with this decision. Therefore, unlike Lemma 2, the product prices and the input of platform services in Table 3 are not related to the value created by the metaverse ecosystem.

Similarly, the findings consistent with Lemma 2 also include the impact of the transfer payment fees on these equilibrium outcomes. As mentioned earlier, the physical brand will increase product prices to cover the direct profit loss caused by the service fee. While the technology subsidy indirectly subsidizes consumers contributing to market knowledge acquisition and application so that product prices will decrease as the subsidy increases. Besides, due to the indirect impact of increased expenditures on consumer demand, that is, in Strategy HS, market demand decreases as the product price increases. If the physical brand does not increase the service input at this time, the demand will be further reduced. While in Strategy HT, the price decreases with the technology subsidy, resulting in demand expansion. The physical brand can control the service input by reducing the investment level. The impact of costs on other outcomes is also consistent with the description in Lemma 2. The intuitive reason is that multiple revenue and cost sources influence the physical brand's and virtual brand's decisions. If more knowledge management contributions brought by the interactive model are needed to offset the costs, these outcomes will decrease with the transfer payment fees.

Numerically, we use p^{HS^*}, p^{HT^*} and a^{HS^*}, a^{HT^*} for simplicity, and the numerical examples with settings $\lambda=0.55, \tau=5, k=26$ *and* $\lambda=0.7$, I: $\tau=1.2, k=12$; II: $\tau=5, k=26$. Other parameters are set to $\gamma=1$ and

Table 3
The impact of related parameters on the optimal solutions in the hybrid strategies.

	p^{HS^*}	p^{HT^*}	s^{HS^*}	s^{HT^*}	a^{HS^*}	a^{HT^*}	$\pi_V^{HS^*}$	$\pi_V^{HT^*}$	$\pi_P^{HS^*}$	$\pi_P^{HT^*}$
γ	*N/A*	*N/A*	*N/A*	*N/A*	↑	↑	↑	↑	↑	↑
τ	*N/A*	*N/A*	*N/A*	*N/A*	↑	↑	↑	↑	↑	↑
e	↑	↓	↑	↓	↕	↕	↕	↕	↕	↕
k	*N/A*	*N/A*	*N/A*	*N/A*	↓	↓	↓	↓	↓	↓
θ	↕	↕	↕	↕	↕	↕	↕	↕	↕	↕

Note: ↑: *indicates increasing*; ↓: *indicates decreasing*; ↕: *indicates non-monotonic*

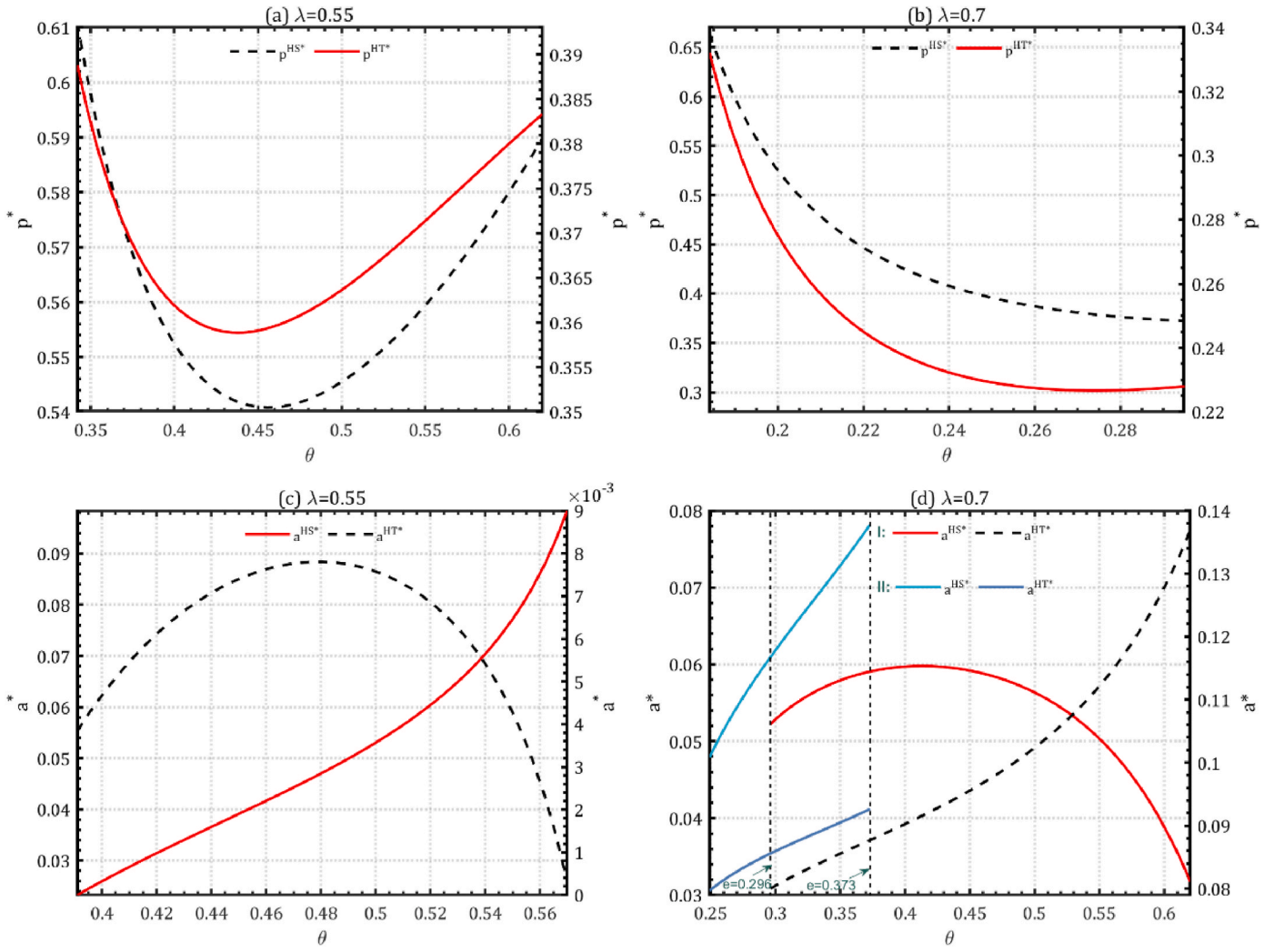


Fig. 7. The impact of parameter θ on the product prices and technology investment levels.

$e = 0.05$. The results are shown in Fig. 7.

Fig. 7(a) and (b) show that when the revenue-sharing ratio is low, the product price will first decrease and then increase as the product fitness increases. When the revenue-sharing ratio is high, the product price will continually decrease. First, a lower revenue-sharing ratio indicates that the physical brand will receive more revenue and gain tremendous advantages. Second, the increase in parameter θ has gradually increased consumers' willingness to pay. When the product fitness is low, the demand is insufficient to offset the profit loss caused by the platform service input. The physical brand will initially decide to charge a higher price to cover costs. As parameter θ increases, the physical brand will appropriately reduce the price to pursue demand expansion. When the product fitness reaches a higher level, both the demand and revenue-sharing ratio can enable the physical brand to obtain higher revenue, and the physical brand is more confident in increasing the price to get more significant profit. On the contrary, when the revenue-sharing ratio is low, the virtual brand is more advantageous, and the physical brand's revenues are less likely to offset the loss. Therefore, the physical brand will always aim to ensure demand, gradually reducing the price as the product fitness increases.

Based on the equilibrium results under the hybrid strategies, we also derive Corollary 2.

Corollary 2. *The equilibrium outcomes in Strategies HS and HT satisfy*:

(i) *The product price and the investment level of platform services in Strategy HS are higher than those in Strategy HT, i.e.,* $p^{HS^*} > p^{HT^*}$ *and* $s^{HS^*} > s^{HT^*}$;

(ii) *When the marginal utility brought by interactive experience transmission* τ *satisfies that* $\tau < 2\gamma\theta/((1-\theta)(2\theta+\lambda-1))$, *then the level of technology investment in Strategy HT is higher than that in Strategy HS, i.e.,* $a^{HT^*} > a^{HS^*}$.

(iii) *The physical brand's profit in Strategy HT is higher than that in Strategy HS, i.e.,* $\pi_P^{HT^*} > \pi_P^{HS^*}$.

(iv) *When the parameters satisfy that* $\tau < 2\gamma\theta/((1-\theta)(2\theta+\lambda-1))$ *and* $k < k_p$, *then the virtual brand's profit in Strategy HT is higher than that in Strategy HS, i.e.,* $\pi_V^{HT^*} > \pi_V^{HS^*}$.

The expression of k_p is defined in Appendix B.3.

Corollary 2 compares equilibrium outcomes under different strategies. We find that the product price and investment level for the brand alliance in Strategy HS will be greater than those in Strategy HT. As mentioned earlier, the physical brand can set a higher price if it pays a service fee and a lower price if it receives a subsidy. In Strategy HS, although a higher price reduces consumers' willingness to pay, the investment level for platform services attracts more consumers to choose the retail model, so more consumers unwilling to buy products will switch to purchasing products through platform services. In the hybrid strategies, consumers' willingness to choose the interactive model depends on the level of platform services s and their technology utility

perceived by knowledge transmission τa gained through the metaverse ecosystem. In most cases, when the technology utility is lower than a certain level, more consumers will be attracted to the retail model under Strategy HS, which will further encourage consumers who want to experience interactive technologies to switch to simple platform services, but the virtual brand's decision level in Strategy HT is still higher, so this also indicates that there will be more consumers using the metaverse ecosystem in Strategy HT than in Strategy HS.

An important finding in Corollary 2 is that the physical brand's profit in Strategy HT will be greater than in Strategy HS. Although the physical brand can obtain higher revenue due to higher total demand in Strategy HS, it loses the advantage due to two service fees. On the one hand, the cost of brand alliance input into Strategy HS is higher due to the high investment level of platform services. On the other hand, the physical brand can obtain more technology subsidies due to more consumers contributing knowledge acquisition and application in Strategy HT. Therefore, the situation in Strategy HT is better than that in Strategy HS for the physical brand. For the virtual brand, Strategy HS is better in most cases. Strategy HT will be favorable for the virtual brand, if and only if the knowledge contribution obtained from adopting the metaverse ecosystem can be fully exploited.

Furthermore, in the hybrid strategies, as product prices and brand alliances' investment levels will show non-monotonic variations and ultimately affect the total market demand and consumers' willingness to use the metaverse ecosystem and platform services, each demand is also affected by multiple factors and will present non-monotonic variations. Therefore, the virtual brand will strategically adjust the level of the technology investment based on these cost payments and revenue earned. Because it is difficult to illustrate this non-monotonic variation case, we use Fig. 7(c) and (d) to depict the impact of the product fitness on the level of technology investment and the high and low situations of the virtual brand's decision level. Fig. 7 indicates that the virtual brand will determine a higher level of technology investment only when the technology cost is not too high. Otherwise, the virtual brand will face a significant cost loss. Moreover, when consumers' technology utility is low, the decision-making level in Strategy HT is higher than that in Strategy HS, consistent with Corollary 2.

However, under different revenue-sharing ratio levels and technology utility, the decision-making levels of the virtual brand show inconsistent variations. The non-monotonic variations are because that the virtual brand's revenues cannot offset the cost losses. The increase in parameter θ will gradually raise the willingness of consumers to use simple platform services so that more consumers who are not willing to buy products or those who want to use the metaverse ecosystem will turn to purchasing products through platform services, which will increase the total demand. The virtual brand has a profit advantage within a specific range. When parameter θ reaches a high level, the total market demand will decrease due to the change in price and investment level for platform services. At this time, the decision will show a downward trend because the revenues obtained making it more challenging to cover cost losses. Corollary 2 shows that when consumers have high technology utility, the virtual brand's profit in Strategy HT is lower than that in Strategy HS, and it is easier to show a downward trend. On the contrary, in the case of low technology utility, the decision-making level under Strategy HT will always increase as parameter θ increases. Fig. 7(d) can also prove that, under the constraints that the numerical system II needed to satisfy, because parameter θ cannot reach a higher value, the decision level of the virtual brand in Strategy HT will maintain an increasing trend as the product fitness increases.

4.2.4. Optimal deployment of knowledge management strategy

Considering that the hybrid strategies have multiple costs, it is necessary to ensure that the equilibrium outcomes of each strategy are non-negative and meaningful. As shown in Appendix A, the pure knowledge management strategy and the hybrid knowledge management strategy have different constraints, and we obtain Propositions 6 and 7 under the constraints of revenue-sharing ratio ($0 < \lambda < 0.6$ and $0.5 < \lambda < 0.7$) and transfer payment fee ($e > \lambda$ and $e < e_4$), respectively. Moreover, Corollary 1 shows that when the physical brand receives the technology subsidy, the virtual brand and the physical brand can obtain more significant performances in most cases. Simultaneously, Corollary 2 also indicates that the physical brand's profit under the technology subsidy strategy will be greater than that under the service fee strategy. Therefore, we further obtain the following insights through the previous theoretical analysis.

Proposition 6. *Based on the optimal profit in Strategies R, IT, and HT, the physical brand's profits in the presence of metaverse ecosystems satisfy*:

(i) *If the technology subsidy satisfies that* $e(\lambda)_1 < e < e(\lambda)_2$, *the physical brand's profit in Strategy R is higher than that in Strategy IT, i.e.,* $\pi_P^{R^*} > \pi_P^{IT^*}$.

(ii) *The physical brand's profits in Strategies R and HT satisfy that* $\pi_P^{HT^*} > \pi_P^{R^*}$; *note that the development cost k needs to satisfy that* $k < k_5$, *then we can obtain the constraint* $e < e_5$.

The expressions of e_5, $e(\lambda)_1$ and $e(\lambda)_2$ are defined in Appendix B.3.

We also explore brand alliances in these knowledge management strategies from the virtual brand's perspective by comparing profits in Strategies R, IT, and HT. The results are shown in Proposition 7.

Propositions 7. *Based on the optimal profit in Strategies R, IT, and HT, the virtual brand's profits in the presence of metaverse ecosystems satisfy*:

(i) *When the parameters satisfy that* $2k(\lambda - 2) + (\gamma + \tau)^2 > 0$, *if and only if the subsidy satisfies that* $e(\lambda)_3 < e < e(\lambda)_4$, *then the virtual brand's profit in Strategy R is higher than that in Strategy IT, i.e.,* $\pi_V^{R^*} > \pi_V^{IT^*}$; *otherwise, when* $2k(\lambda - 2) + (\gamma + \tau)^2 < 0$, *the virtual brand's profit in Strategy IT is higher than that in Strategy R if* $e(\lambda)_3 < e < e(\lambda)_4$, *i.e.,* $\pi_V^{IT^*} > \pi_V^{R^*}$.

(ii) *When the parameters satisfy that* $E > 0$, *if and only if the subsidy satisfies that* $e(\lambda)_5 < e < e(\lambda)_6$, *then the virtual brand's profit in Strategy R is higher than that in Strategy HT, i.e.,* $\pi_V^{R^*} > \pi_V^{HT^*}$; *otherwise, when* $E < 0$, *the virtual brand's profit in Strategy HT is higher than that in Strategy R if* $e(\lambda)_5 < e < e(\lambda)_6$, *i.e.,* $\pi_V^{HT^*} > \pi_V^{R^*}$.

The expressions of E, $e(\lambda)_3$, $e(\lambda)_4$, $e(\lambda)_5$ and $e(\lambda)_6$ are defined in Appendix B.3.

Numerically, we set other related parameters to $\gamma = 1$ and $\theta = 0.5$, and ensure that the above parameters satisfy their various constraints to get our insights. Propositions 6 and 7 depict the optimal deployment of knowledge management strategies when considering two brands' profits. First, by comparing the profits under different strategies, we get optimal decisions focusing on the revenue-sharing ratio and the technology subsidy, as shown in Figs. 8 and 9. Specifically, when the revenue-sharing ratio is at a low level, and the technology subsidy of the virtual brand is high, both the physical brand and the virtual brand should choose the pure strategy, that is, the retail model under brand integration or the interactive model in the metaverse ecosystem. However, when the subsidy is low, and the revenue-sharing ratio exceeds the medium level, it is always optimal for the physical brand to choose a hybrid knowledge management strategy, and the virtual brand should also choose between the hybrid strategy and the pure retail strategy.

Moreover, in most cases, it is optimal for the physical brand to choose a pure metaverse strategy or a hybrid metaverse strategy. On the one hand, the physical brand can directly benefit from the comprehensive knowledge management brought by virtual-physical alliances and obtain certain technology subsidies. On the other hand, under the pure retail strategy, the physical brand has to pay the corresponding service input for the brand integration to meet the increased demand. If and

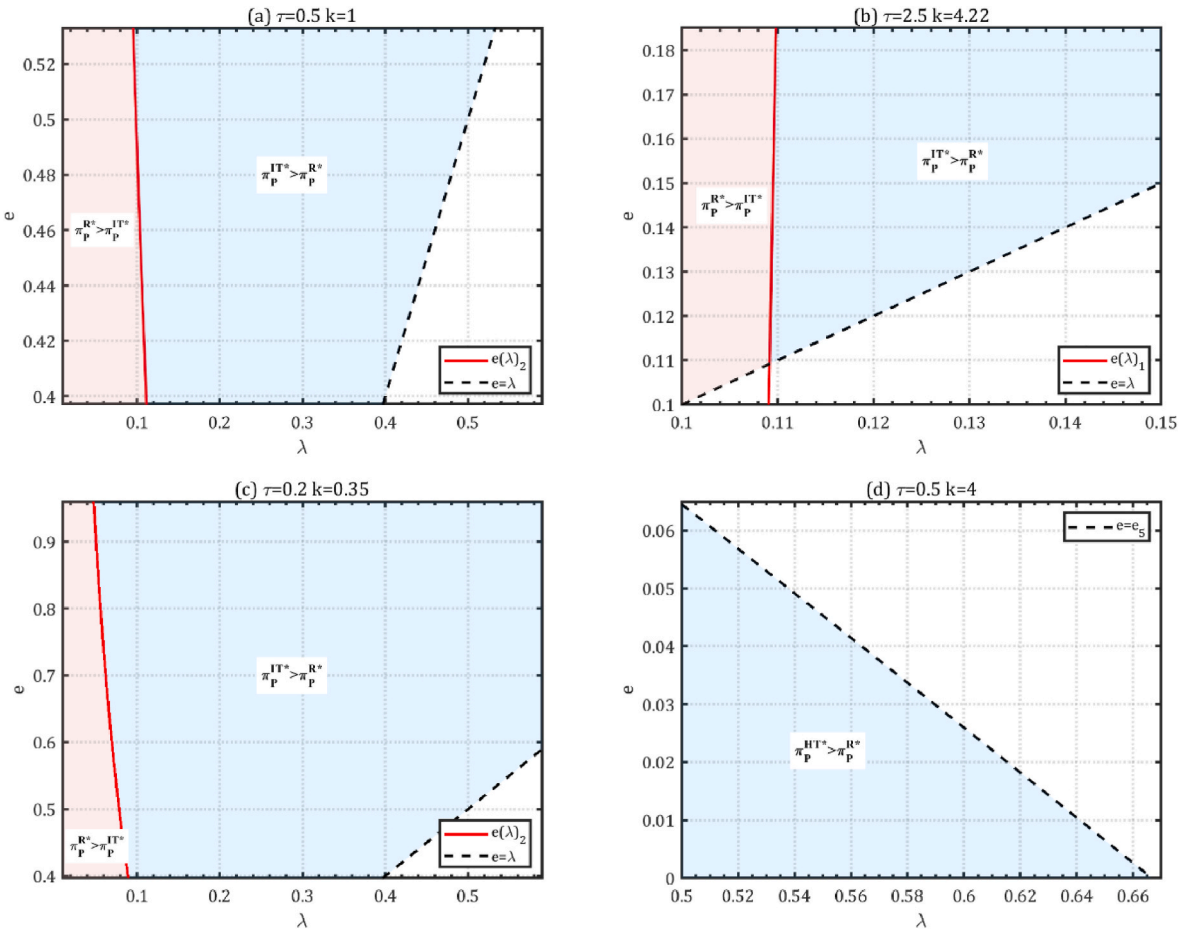


Fig. 8. The physical brand's optimal deployment in the presence of metaverse ecosystems.

only if the revenue-sharing ratio determined by the virtual brand is low, the physical brand will choose the retail model because, at this point, the physical brand can obtain a more significant portion of the revenue and is in a favorable position under the revenue-sharing contract. On the contrary, for the virtual brand, the retail model is optimal in most cases. According to Corollary 1, when consumers' technology utility perceived by knowledge transmission is high due to the increase in technology cost, the virtual brand's profit under Strategy IT will be at a lower level. If the revenue-sharing ratio level is still low, the virtual brand will not choose the metaverse ecosystem strategy, which is inevitable. And Corollary 2 also proves that only when the technology utility and technology cost are low, in a hybrid strategy, the virtual brand chooses a technology subsidy instead of a service fee to be more optimal. Therefore, if the technology utility is higher, the physical brand cannot consider metaverse ecosystems in a pure or hybrid strategy, as they cannot achieve a win-win situation between the physical and virtual brands. In other cases, no matter which strategy the alliance chooses, it can always achieve the Pareto optimum among two members.

4.3. Strategy selection and case study

4.3.1. Case description

Recently, many luxury and sports brands have collaborated with virtual brands or metaverse platforms to launch their own virtual modules, selling their fashion items or bundling virtual and physical products for sale (Ahn et al., 2024; Zhang & Ye, 2023). The metaverse creates a new business model of virtual-physical alliances between brands, including the direct retail model using the virtual brand's existing platform and the immersive interactive experience model within the metaverse ecosystem. These models help physical brands, especially start-up brands, expand their markets, and bundling sales with physical products can also create sustainable performance for virtual brands. Following the research of Mancuso et al. (2024), we also select the Roblox metaverse platform as our research object. As the world's leading metaverse platform, Roblox can create metaverse ecosystems and a huge user base, occupies a leading position in the metaverse market, and is committed to providing metaverse technology services and solutions for relevant enterprises. With the development of metaverse technologies, the platform collaborates with many well-known traditional brands and enterprises for metaverse business practices, especially the new business model of alliance with sports fashion brands.

The choice of the physical brand Nike is based on it becoming one of the first brands to establish a high position in the metaverse. Nike is brave in exploration and innovation and successfully built the immersive metaverse virtual space Nikeland, leading the metaverse business practices of other brands. Nike partnered with the platform Roblox to launch Nikeland in September 2021 and sells sneakers and clothes in Roblox's Nikeland. Nikeland integrates multiple models, such as gaming, socializing, and shopping. Consumers can hang out, shop inside, and chat with strangers. To encourage movement, visitors to Nikeland can use accelerometers on their mobile devices to translate offline movement into online play. For example, players can move their device and body in real life to execute in-game moves like long jumps or speed runs. In Nikeland's showroom module, consumers can outfit their virtual avatars with Nike's shoes and clothes for individual preference. Fig. 10 depicts the relevant themes and panorama of Nikeland.

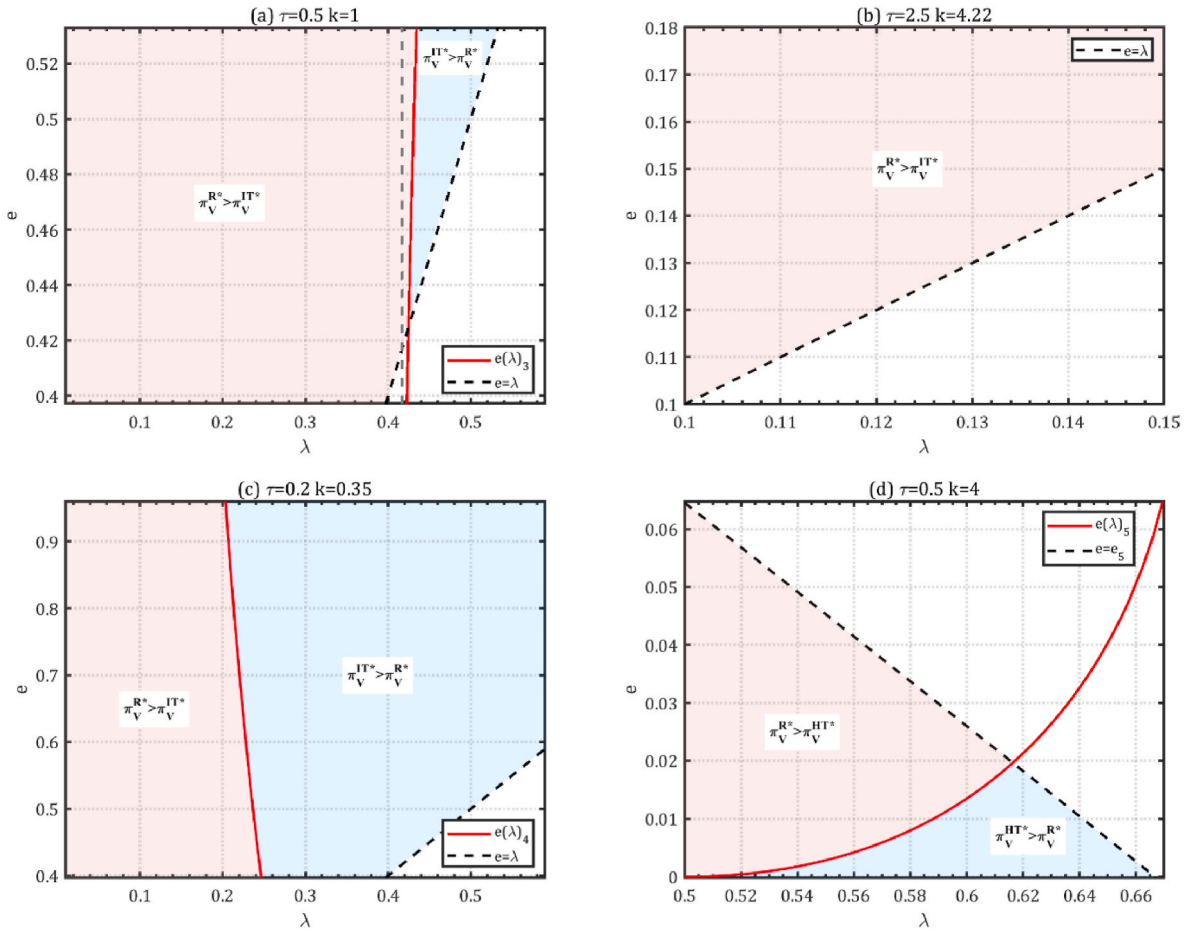


Fig. 9. The optimal deployment from the perspective of the virtual brand.



Fig. 10. Nikeland courtesy of Nike.

Table 4
Values of the parameters are used to illustrate the optimal deployment decisions.

Parameter	Definition	Value	Source
θ	Fitness of products without interactive knowledge transmission	0.5	Sun et al. (2022)
τ	Marginal utility brought by interactive experience knowledge transmission	1.02	The direct sales value of technology utility
γ	Marginal value that market knowledge contributed in the metaverse ecosystem	3	Sun and Ji (2022)
k	Technology development cost of the metaverse ecosystem in the pure strategy	4.9	The values that satisfy the constraints
	Technology development cost of the metaverse ecosystem in the hybrid strategy	15	

4.3.2. Data collection

For the case data of Roblox and Nike, we use second-hand data obtained from a relevant website (snobette.com, 2021) and literature sources. We use available numerical values from them as the values for the key parameters of our model. First, we collected sales data directly related to the case, and then we supplemented the values of other parameters from previous literature. The values of related parameters are summarized in Table 4. In our case, Roblox and Nike formed a strategic alliance where consumers can purchase virtual products in their collaborative metaverse, Nikeland. According to Roblox's report data released by the site, in the same year after the launch of Nikeland, Roblox reported third-quarter revenues of $509.3 million, up to 102 percent over the prior year. The stock jumped 33 percent on the news. Based on this data, we can assume that the profit growth is the direct sales value of metaverse technologies' utility, so the technology utility τ is set to 1.02.

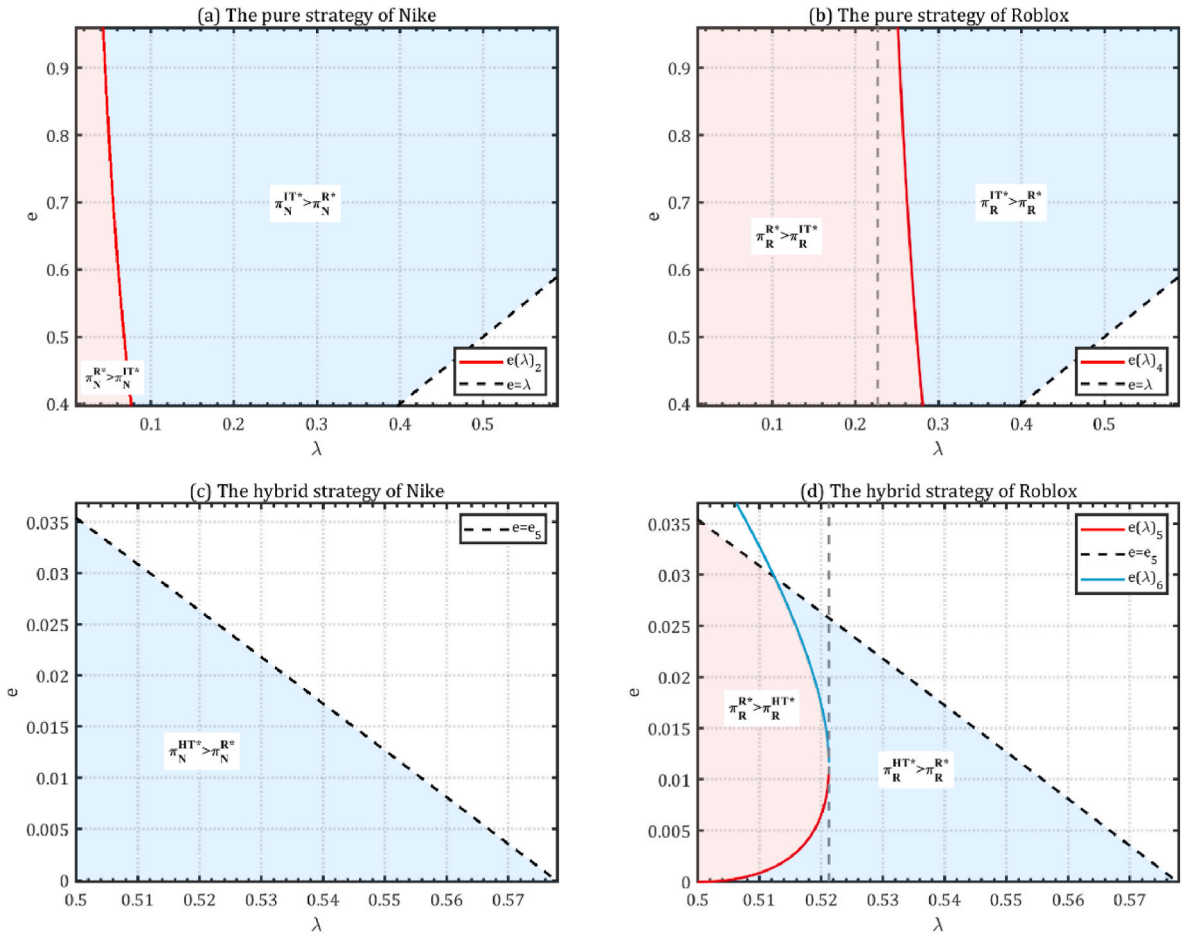


Fig. 11. The optimal deployment decisions for Nike and Roblox.

4.3.3. Results

By solving the equilibrium outcomes of the model, we can obtain the comparisons of Roblox's and Nike's profits in Strategies R, IS, IT, HS, and HT, and the results are as follows:

Nike's profit in Strategy IT will be higher than that in Strategy IS, i.e., $\pi_N^{IT*} > \pi_N^{IS*}$, which is consistent with Corollary 1. Besides, Nike's profits in Strategies R, HS, and HT also satisfy that $\pi_N^{HT*} > \pi_N^{HS*}$ and $\pi_N^{HT*} > \pi_N^{R*}$, in line with Corollary 2 and Proposition 6. Finally, we also determine feasible regions for both enterprises to choose knowledge management strategies, including alliance selection of pure knowledge management strategies and hybrid knowledge management strategies, as shown in Fig. 11. In the partnership established between Roblox and Nike, Roblox provides the metaverse market and technology services to Nike. Nike needs to pay a certain proportion of revenue-sharing as an admission fee and also a technology service fee or obtain a subsidy. Under the control of revenue and expenditure costs, the optimal strategy choice of the alliance is determined. As shown in Fig. 11, Nike and Roblox can achieve a win-win situation through strategic alliances.

5. Discussion

The metaverse has strategic significance for economic and social development. This study extends the existing body of literature by moving beyond conceptual understandings and general attitudes toward the metaverse, delving into its impact on knowledge-driven business model innovation. Although many studies have examined the potential of the metaverse for business model transformation such as marketing (Bruni et al., 2025; Buechel & Spinler, 2024), services (Huynh-The et al., 2023), and hotel tourism (Gursoy et al., 2022), this paper explores the internal logic behind the emergence of new business models in the metaverse. The results provide support for knowledge management theories and practices that integrate the physical and virtual economies.

We develop a knowledge management framework for the metaverse virtual-physical alliance, where virtual brands need to know the professional expertise of physical brands in product design, including product content and design details. Physical brands can obtain virtual brands' marketing channel resources utilizing platform channels or metaverse ecosystems. By integrating professional knowledge, the two can jointly launch virtual products derived from physical products or virtual physical-bundled products (Ahn et al., 2024; Yoo et al., 2023; Zhang & Ye, 2023). With the alliance of physical brands, virtual brands can achieve sustainable sales performance, acquire and transmit two-way customer market knowledge, create additional customer interaction value, and create conditions for physical brands entering the metaverse to expand global markets and enhance brand influence.

For practitioners in the metaverse market, the framework offers actionable insights into the deployment of knowledge management strategies to enhance customer value and economic sustainability. Interactive models in the metaverse ecosystem can provide consumers with an enhanced technology experience and create additional customer value. They overcome the limitations of solely improving performance through brand knowledge integration in retail models (Hoyer et al., 2020; Lemon & Verhoef, 2016; Sun & Ji, 2022), thus helping brands to tailor their adoption strategies effectively, ensuring a win-win scenario that maximizes profitability and customer satisfaction.

Our study explores the impact of key parameters on brands' strategic decisions, providing decision support in terms of pricing, technology investment, cost control, and market strategies. Some parameters can

affect the pricing decisions of the physical brand by influencing consumers' purchasing demand, further affecting brands' equilibrium profit. The research results provide theoretical extensions for literature in the field of general management, such as strategic management, technological innovation, and service marketing (Ancillai et al., 2023; Chakraborty et al., 2024; Jafar et al., 2023; Yan et al., 2022), and also help practitioners and policymakers make optimal decisions strategically by weighing value creation and technology investment when facing market uncertainty.

Our results align with and build upon the fundamental work of Sun and Ji (2022) and Yan and Han (2022) on information technologies' economic impact and the relevant theories of customer knowledge management (Abbas & Khan, 2023; Aghamirian et al., 2015; Schniederjans et al., 2020). By introducing the two-way value creation brought by customer knowledge acquisition and transmission—encompassing both enhanced consumer utility through improved technology experience transmission and created customer engagement value via market knowledge acquisition—technology utility can expand the market by increasing consumers' willingness to pay, and customer value can benefit brands through direct contributions.

Based on Chen, Ye, & Huang's (2023), Wang et al.'s (2022), and Abdulmuhsin et al.'s (2024) results on metaverse technologies reshaping knowledge management processes in related fields, our results explore the impact of knowledge management practices improved by metaverse technologies on brand decision-making. These insights expand theoretical insights into the interplay between (metaverse) technological innovation and customer knowledge management, and the field of metaverse service marketing (Ahn et al., 2024; Bag et al., 2021; Centobelli & Ndou, 2019; Pillai et al., 2023). Simultaneously, this requires enterprises to effectively use the relationship between technology and knowledge management and attach importance to adopting new market strategies and value propositions, thereby playing the role of consumers in value creation.

The research also examines the impact of strategic interaction choices in virtual-physical alliances on the brands' optimal decisions. The virtual and physical brands can determine the ways of value creation and acquisition through strategic alliances, thus realizing new forms of economic transactions with partners, which is an important part of business model transformation (Mancuso et al., 2024; Ramon Saura et al., 2023; Scuotto et al., 2022; Yi et al., 2020). Brands reach supply chain alliances through revenue sharing and transfer payments and integrate and apply each other's complementary knowledge in knowledge management practice to form a value collaboration network. We examine the important role of value collaboration networks realized by new economic transactions from the combined angle of knowledge management and strategic alliances and analyze the impact of strategic dynamics on brands' strategic decisions from a subtle perspective. The relationship networks formed by strategic alliances help enterprises to expand new businesses and markets in a wider range of knowledge sources through a more stable structure, coping with various business challenges (Zahoor et al., 2024; Loorbach & Wijsman, 2013; von Delft et al., 2019; Martins et al., 2019). This fusion of virtual and physical is essential for virtual brands to achieve long-term sustainable performance and to help physical brands remain unique in the face of increasingly global competition.

From a more subtle perspective of strategic decision-making, following Sun and Ji (2022) and Wu and Wang (2023), we examine the impact of transfer payment fees in a theoretical model. The increase in virtual brand charges causes the physical brand to raise the prices to offset the losses, while the technology subsidy indirectly subsidizes consumers who contribute to the acquisition and transmission of market knowledge, making brand pricing lower than the general level. At this point, technology subsidy strategies yield higher profits for brands by expanding market demand and leveraging customer engagement (Sun, 2012; Yan & Han, 2022). This also aligns with the sustainable business practices advocated by digital transformation scholars, especially the role of innovation from markets and business model transformation (Ancillai et al., 2023; Barham et al., 2020; Cresnar et al., 2023; Sjodin et al., 2020). The strategic interactions under brand alliances allow brands to enhance consumer accessibility through technology subsidies, addressing social sustainability concerns such as the digital divide and inclusion. By balancing strategic alliance dynamics with customer engagement initiatives, firms can achieve a more sustainable business ecosystem.

We ultimately examine the optimized deployment of brands' knowledge management practices, i.e., the optimal choice for virtual-physical alliance business models with different knowledge management strategies. Tomini et al.'s (2024), Zabel et al.'s (2023), and Mancuso et al.'s (2024) work further proved that business model innovation requires the effective management of various resources and elements, including aspects such as technology, knowledge management, and stakeholder dynamics. However, it is still a need to explore the relationships among the constituent elements of metaverse business ecosystems from a multi-integrated angle to provide practical guidance for organizational business model innovation. The evolution of the business model in the metaverse ecology is essentially due to the improvement of the knowledge management process relying on different relationship networks and different degrees of technological change in the metaverse, resulting in business models evolving in different forms in the metaverse. The comprehensive perspective we propose unifies previous studies on the interplay between the metaverse, knowledge management, and business model innovation (Abdulmuhsin et al., 2024; Ancillai et al., 2023; Buechel & Spinler, 2024; Hadi et al., 2024) to explore optimal strategic decisions for brands from a subtle model equilibrium analysis. This study provides practitioners and policymakers with a correct understanding and utilization of key elements such as technology, knowledge, and relationship networks to help enterprises successfully transform their business models.

In summary, this discussion reinforces the crucial role of metaverse ecosystems in leveraging knowledge management strategies to drive business model evolution. By bridging empirical findings with existing literature, the study not only validates theoretical propositions but also extends them for understanding and utilizing the inseparable relationships between the metaverse, knowledge management, and business model innovation. Discussions of management implications provide a strategic blueprint for metaverse service providers and brands, emphasizing the importance of knowledge management practices and strategic investments in technology to foster sustainable growth and competitive advantage.

Table 5
The list of theoretical contributions and practical implications of this study.

Main findings	Theoretical contributions	Practical implications
Business Model Evolutions under the Ternary Relationship	Conceptual Framework of Knowledge Management	Comprehensive Knowledge Management Practices
	Elements of Business Model Innovation	Business Model Innovation Guidance
Strategic Dynamics and Knowledge Performance	Knowledge Management and Value Co-creation	Brand Strategy Interaction and Knowledge Collaboration
		Market Strategy and Customer Knowledge Value
Transfer Payment Options for Market Participation	Knowledge Management and Value Co-creation	Brand Strategy Interaction and Knowledge Collaboration
		Market Strategy and Customer Knowledge Value
Optimal Knowledge Management Strategy Deployment	Empirical Support for Theoretical Propositions	Knowledge Ecosystem Optimization and Sustainable Strategy

6. Conclusion

This study examines the knowledge management practices based on different market forces in the metaverse business ecosystem. Those are the integration, acquisition, transmission, and application of knowledge under virtual-physical alliances as internal driving knowledge elements to reconstruct the existing business model of enterprises. We explore the business evolution logic driven by knowledge management, and the sustainable performance brought about by its value chain transmission. Brands improve their knowledge management processes through metaverse technologies to maintain external relationship networks, including strategic alliances and customer participations, forming a collaborative network for value creation, acquisition, and transmission, enabling successful business model transformation, which has important implications for practitioners and policymakers to effectively harness the critical role of technological innovation, knowledge management, and relationship networks. For clearer presentations, we summarize the main findings and corresponding contributions of this paper in Table 5.

6.1. Main findings

This paper examines the tradeoffs between brand and physical brands in knowledge management strategies, including retail models under brand knowledge integration, interactive models in virtual ecosystems, and hybrid knowledge management strategies of the two models. By comparing the total profits of virtual and physical brands under different alliance models, the optimal deployment of brands' knowledge management strategy is determined. Utilizing a mixed-methods approach, including quantitative data analysis and qualitative case studies from leading metaverse platform entities, the study identified several key findings.

(1) Business Model Evolutions under the Ternary Relationship: The metaverse technologies improve knowledge management processes, maintaining good interactive relationships between strategic alliance partners, as well as between enterprises, customers, and consumers. These relationships further promote the participation of these stakeholders in metaverse ecosystems. These knowledge management processes rely on different relationship networks and technological changes, including knowledge integration, application, acquisition, and transmission, enabling business models to evolve synchronously according to different ways of value creation, acquisition, and transmission.
(2) Strategic Dynamics and Knowledge Performance: The value collaboration network relying on virtual-physical alliances promotes the value co-creation among partners through the strategic structure of knowledge complementarities and integrations. In interactive customer engagement, the metaverse ecosystem significantly enhances consumer utility by improving technological experience and knowledge acquisition and transmission through market interaction, which can generate spillover effects of customer engagement, creating additional customer value for the brand. Through equilibrium analyses of key parameters, stakeholders' strategic dynamics affect the sales performance of brand alliances by influencing decisions such as pricing, technology investment, and market strategies.
(3) Transfer Payment Options for Market Participation: The adoption of the metaverse ecosystem can benefit service providers from customer knowledge contributions. The analysis of transfer payment fees under brand alliances reveals that technology subsidies generally yield higher profits for both virtual and physical brands by expanding market demand and leveraging customer engagement.
(4) Optimal Knowledge Management Strategy Deployment: The study determined that, under most conditions, adopting an interactive model in the metaverse ecosystem is more optimal, and providing technology subsidies is more beneficial than charging service fees. This strategy fosters a win-win scenario by maximizing profitability and customer satisfaction, thereby achieving a Pareto optimum in virtual-physical alliances even when the technology utility is low.

6.2. Theoretical contributions and practical implications

The theoretical contributions of this study are as follows.

(1) Knowledge Management Conceptual Framework: We construct a conceptual framework for knowledge management regarding different market forces, including knowledge acquisition, integration, transmission, and application. This framework contributes to the theoretical research of business model innovation, especially the new perspective of examining the ternary relationships between the metaverse, knowledge management, and business model innovation, which helps the new research regarding business model innovation to locate so that their research findings can be discussed according to different perspectives and new understandings.
(2) Elements of Business Model Innovation: We introduce the evolution of business models in the influence of the metaverse, i.e., from traditional e-commerce models to retail models that integrate brand knowledge, to interactive models in the metaverse ecosystem. The role of metaverse technologies, knowledge management, and good relationship networks between alliance partners and customers we explore also provides more subtle ideas for research in the field of business model innovation.
(3) Knowledge Management and Value Co-creation: Our study enriches the literature on the intersection of knowledge management and strategic alliances, as well as the field of customer knowledge management. The integration and application of knowledge under strategic alliances form a value collaboration network, and the acquisition and transmission of customer market knowledge through interactive customer participation create a dual knowledge contribution of direct and indirect value for brands. By optimizing the deployment of knowledge management strategies, value co-creation can be achieved.
(4) Empirical Support for Theoretical Propositions: The equilibrium analyses in theoretical propositions explore the impact of key parameters on brands' strategic decisions and provide decision support from pricing, technology investment, cost control and marketing strategies. The findings provide theoretical extensions to the literature in the fields of general management such as strategic management, technological innovation, and service marketing. Simultaneously, the numerical analysis with the case study provides support for the rationality of our theoretical insights on strategic optimization deployment.

There are also several important implications for practitioners.

(1) Comprehensive Knowledge Management Practices: Comprehensive knowledge management practices enable the evolution of brand business models, which integrate and apply complementary knowledge between virtual-physical alliance partners, acquiring and transmitting customer market knowledge interactively. By improving knowledge management processes to generate new ways of value creation, acquisition, and transmission, enterprises can leverage (metaverse) technological change to transform the existing business landscape.
(2) Business Model Innovation Guidance: Our comprehensive framework reflects the mechanism of action between metaverse technology, knowledge management, and business model innovation. Practitioners and management decision-makers can design specific business models based on key elements and

research objects, including economic transactions between strategic alliance partners and market strategies for new products and services. The framework provides reference and practical guidance for enterprises' digital transformation of business models and the metaverse business practices.

(3) Brand Strategy Interaction and Knowledge Collaboration: The value collaboration network formed by strategic alliances helps enterprises expand new businesses and markets through a more stable structure and a wider range of knowledge sources and respond to various business challenges. The strategic interaction under virtual-physical alliances enables brands to improve consumer accessibility through technology subsidies, addressing social sustainability concerns such as the digital divide and inclusion.

(4) Market Strategy and Customer Knowledge Value: The interactive models in the metaverse ecosystem can provide consumers with an enhanced technology experience and create additional customer value, overcoming the limitations of solely improving performance through brand knowledge integration in retail models. This requires enterprises to effectively utilize the relationship between technologies and knowledge management and attach importance to adopting new market strategies and value propositions, thus leveraging the role of consumer participation in value creation.

(5) Knowledge Ecosystem Optimization and Sustainable Strategy: Implementing robust knowledge ecosystems enable brands to effectively acquire and transmit consumer market knowledge, informing strategic resources for brand operations. Subtle optimization decisions, including product pricing, technology investment, and market participation, help practitioners and policymakers achieve a more sustainable business ecosystem by balancing strategic alliance dynamics with customer engagement initiatives when facing market uncertainty.

6.3. Limitations and future research

While this study provides valuable insights, it has several limitations that warrant further investigation.

(1) Limited Strategic Alliance Scenarios: This study explores the deployment of knowledge management strategies through a single physical brand and a single virtual brand. However, in more complex business ecosystems, the theoretical framework of knowledge needs further discussion, and future research should consider more complex commercial structures to provide a more comprehensive understanding.

(2) Customer Engagement Assumption: Assuming that all consumer's engagement in the market is through a single alliance strategy, which may oversimplify interaction with the market and limit the applicability of the results. Future studies can examine the behaviors of multi-channel or repeated engagement in the market and their implications on acquiring customer knowledge performance.

(3) Consumer Segmentation: Our theoretical model assumes that consumers strategically choose the sales channels provided by the brand through maximizing utility. Future research could investigate consumer segments indifferent towards technological experiences, identifying barriers to metaverse-driven engagement and informing more inclusive digital strategies.

(4) Cross-Disciplinary Approaches: Integrating insights from management, international business, geopolitics, MIS, and entrepreneurship can provide a holistic understanding of the metaverse's impact on socio-economic sustainability. Cross-disciplinary studies are encouraged to explore the multifaceted effects of these technologies.

In conclusion, this study emphasizes the construction of the metaverse's commercial ecosystem and value chain, i.e., the brand strategy of virtual-physical alliances, and captures the potential of metaverse technology in utilizing knowledge management strategies to drive global business model transformation. By bridging empirical findings with existing literature, the study not only validates theoretical propositions but also extends them for understanding and utilizing the inseparable relationships between the metaverse, knowledge management, and business model innovation. The practical implications provide a strategic blueprint for metaverse service providers and brands, emphasizing the importance of knowledge management practices and strategic investments in technology to foster sustainable growth and competitive advantage. Future research should continue to explore these dynamics, particularly in the context of emerging technologies and business model evolution, to further advance the field of Technology in Society.

CRediT authorship contribution statement

Baogui Xin: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Yaping Song:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Wei Peng:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Hui Tan:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Consent

All authors have approved the manuscript.

Data transparency

All data support published claims.

Funding

This work was supported by the National Fund of Philosophy and Social Science of China (Grant No. 24AJY030).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Parameter constraints

A.1 Constraints from the case with retail model

We can obtain the Hessian matrix of Strategy R:

$$H_1=\begin{bmatrix}\frac{\partial^2\pi_P}{\partial p^2} & \frac{\partial^2\pi_P}{\partial p\partial s}\\ \frac{\partial^2\pi_P}{\partial s\partial p} & \frac{\partial^2\pi_P}{\partial s^2}\end{bmatrix}=\begin{bmatrix}-\frac{2(1-\lambda)}{\theta} & \frac{1-\lambda}{\theta}\\ \frac{1-\lambda}{\theta} & -1\end{bmatrix}=\frac{(1-\lambda)(-1+2\theta+\lambda)}{\theta^2}, \tag{A.1}$$

then

$$\frac{1-\lambda}{2}<\theta<1. \tag{A.2}$$

A.2 constraints form the case with interactive model

we can obtain the Hessian matrices of both Strategies IS and IT led by the virtual brand:

$$H_2=H_3=\begin{bmatrix}\frac{\partial^2\pi_P}{\partial p^2} & \frac{\partial^2\pi_P}{\partial p\partial a}\\ \frac{\partial^2\pi_V}{\partial a\partial p} & \frac{\partial^2\pi_V}{\partial a^2}\end{bmatrix}=\begin{bmatrix}-2+2\lambda & (1-\lambda)\tau\\ -\gamma+\lambda\tau & -k+2\gamma\tau\end{bmatrix}=(1-\lambda)(2k-\tau(3\gamma+\lambda\tau)), \tag{A.3}$$

then

$$k>\frac{1}{2}(3\gamma\tau+\lambda\tau^2), \tag{A.4}$$

and from the demand and profits $D^{IS^*},\pi_V^{IS^*},\pi_P^{IS^*}>0$ with $D^{IS^*},\pi_V^{IS^*},\pi_P^{IS^*}$, we have:

- $0<e\le 1-\lambda\&k>\frac{1}{2}(3\gamma\tau+\lambda\tau^2)\&\tau>0$,
- $e>1-\lambda\&\tau>\tau_1\&\frac{1}{2}(3\gamma\tau+\lambda\tau^2)<k<k_1$.

We also derive the constraints from the demand and profits $D^{IT^*},\pi_V^{IT^*},\pi_P^{IT^*}>0$ with $D^{IT^*},\pi_V^{IT^*},\pi_P^{IT^*}$:

- $0<\mathrm{e}\le\frac{\lambda^2-\lambda}{\lambda-2}\&\mathrm{k}>\frac{1}{2}(3\gamma\tau+\lambda\tau^2)\&\tau>0$,
- $\frac{\lambda^2-\lambda}{\lambda-2}<\mathrm{e}<\frac{5\lambda^2-5\lambda}{5\lambda-6}\&\frac{1}{2}(3\gamma\tau+\lambda\tau^2)<\mathrm{k}<\mathrm{k}_2\&\tau>0$,
- $e=\frac{5\lambda^2-5\lambda}{5\lambda-6}\&\tau\ne\tau_2\&\frac{1}{2}(3\gamma\tau+\lambda\tau^2)<\mathrm{k}<\mathrm{k}_2$,
- $\frac{5\lambda^2-5\lambda}{5\lambda-6}<\mathrm{e}<\lambda\&(0<\tau<\tau_2\|\tau>\tau_3)\&\frac{1}{2}(3\gamma\tau+\lambda\tau^2)<\mathrm{k}<\mathrm{k}_2$,
- $e=\lambda\&0<\tau<\frac{\gamma(\lambda-1-e)}{3(1-\lambda)\lambda+e(3\lambda-4)}\&\frac{1}{2}(3\gamma\tau+\lambda\tau^2)<k<k_2$,
- $e>\lambda\&0<\tau<\tau_2\&\frac{1}{2}(3\gamma\tau+\lambda\tau^2)<k<k_2$.

To consider the constraints above, we can derive:
when $0<\lambda<\frac{1}{2}$, then $\frac{\lambda^2-\lambda}{\lambda-2}<\frac{5\lambda^2-5\lambda}{5\lambda-6}<\lambda<1-\lambda$,
when $\frac{1}{2}<\lambda<\frac{3}{5}$, then $\frac{\lambda^2-\lambda}{\lambda-2}<\frac{5\lambda^2-5\lambda}{5\lambda-6}<1-\lambda<\lambda$,
when $\frac{3}{5}<\lambda<1$, then $\frac{\lambda^2-\lambda}{\lambda-2}<1-\lambda<\frac{5\lambda^2-5\lambda}{5\lambda-6}<\lambda$,
for simplicity, the constraint $e>\lambda$ is considered in our study, we have:

- if $0<\lambda\le\frac{1}{2}\&\lambda<e\le 1-\lambda$, then $0<\tau<\tau_2\&\frac{1}{2}(3\gamma\tau+\lambda\tau^2)<k<k_2$,
- if $0<\lambda\le\frac{1}{2}\&\lambda<1-\lambda<e$, then $\tau_1<\tau<\tau_2\&\frac{1}{2}(3\gamma\tau+\lambda\tau^2)<k<\min\{k_1,k_2\}$,
- if $\frac{1}{2}<\lambda<\frac{3}{5}\&1-\lambda<\lambda<e$, then $\tau_1<\tau<\tau_2\&\frac{1}{2}(3\gamma\tau+\lambda\tau^2)<k<\min\{k_1,k_2\}$,
- if $\frac{3}{5}<\lambda<1\&1-\lambda<\lambda<e$, then $\tau_1>\tau_2$, there are no reasonable constraints at this point.

A.3 Constraints from the hybrid strategy

We can obtain the Hessian matrices of both Strategies HS and HT led by the virtual brand:

$$H_4=H_5=\begin{bmatrix}\frac{\partial^2\pi_P}{\partial p^2} & \frac{\partial^2\pi_P}{\partial p\partial s} & \frac{\partial^2\pi_P}{\partial p\partial a}\\ \frac{\partial^2\pi_P}{\partial s\partial p} & \frac{\partial^2\pi_P}{\partial s^2} & \frac{\partial^2\pi_P}{\partial s\partial a}\\ \frac{\partial^2\pi_V}{\partial a\partial p} & \frac{\partial^2\pi_V}{\partial a\partial s} & \frac{\partial^2\pi_V}{\partial a^2}\end{bmatrix}=\begin{bmatrix}\frac{2(-1+\lambda)}{\theta} & \frac{1-\lambda}{\theta} & 0\\ \frac{1-\lambda}{\theta} & -1 & 0\\ 0 & -\frac{\gamma}{1-\theta} & -k+\frac{2\gamma\tau}{1-\theta}\end{bmatrix}=\frac{(1-\lambda)(-1+2\theta+\lambda)(k(\theta-1)+2\gamma\tau)}{(1-\theta)\theta^2}, \tag{A.5}$$

then

$$\frac{1-\lambda}{2} < \theta < 1 \& k > \frac{2\gamma\tau}{1-\theta} \tag{A.6}$$

and from the constraint (A.6) and demand as well as the level of technology investment $D_I^{HS^*}, D_R^{HS^*}, a^{HS^*} > 0$ with $D_I^{HS^*}, D_R^{HS^*}, a^{HS^*}$, we can obtain:

- $1-2\theta < \lambda \le 1-2\theta+2\theta^2 \& 0 < \gamma < \gamma_1 \& k_3 < k < k_4$,
- $1-2\theta+2\theta^2 < \lambda \le 1-\theta+\theta^2 \& ((0 < e \le e_1 \& \gamma > 0 \& k > k_3) \| (e > e_1 \& 0 < \gamma < \gamma_1 \& k_3 < k < k_4))$,
- $1-\theta+\theta^2 < \lambda < 1 \& ((e_2 < e \le e_1 \& \gamma > 0 \& k > k_3) \| (e > e_1 \& 0 < \gamma < \gamma_1 \& k_3 < k < k_4))$.

We also obtain the constraints from the retail price and the investment of the physical brand for knowledge integration $p^{HT^*}, s^{HT^*} > 0$ with p^{HT^*}, s^{HT^*}:

$$0 < e < \frac{1}{2}(1-\theta-\lambda+\theta\lambda) \& 0 < e < \theta-\theta^2 \tag{A.7}$$

and from the above constraint (A.7) and the demand as well as the level of technology investment $D_R^{HT^*}, D_I^{HT^*}, a^{HT^*} > 0$ with $D_R^{HT^*}, D_I^{HT^*}, a^{HT^*} > 0$, we have:

- $1-2\theta < \lambda \le 1-2\theta+2\theta^2 \& e_3 < e < e_4 \& \gamma > \gamma_2 \& k > k_5$,
- $1-2\theta+2\theta^2 < \lambda < 1-\theta+\theta^2 \& 0 < e < e_4 \& \gamma > \gamma_2 \& k > k_5$.

Therefore, the constraint $1-2\theta < \lambda < 1-\theta+\theta^2$ needs to be considered in our study, we have:

- if $1-2\theta < \lambda \le 1-2\theta+2\theta^2$, then $\gamma_1 < \gamma_2$, there are no reasonable constraints at this point,
- if $1-2\theta+2\theta^2 < \lambda < 1-\theta+\theta^2$, then $0 < e < \min\{e_1, e_4\} \& \gamma > \gamma_2 \& k > \max\{k_3, k_5\}$ or $e_1 < e < e_4 \& \gamma_2 < \gamma < \gamma_1 \& \max\{k_3, k_5\} < k < k_4$.

A.4 Summary of constraints

We can verify the numerical relationships of the constraints as follows.

- $\theta > \frac{1-\lambda}{2}$ In the case with retail model and in the hybrid strategy.
- $0 < \lambda \le \frac{3}{5}$ In the case with interactive model.
  - (i) if $0 < \lambda \le \frac{1}{2} \& \lambda < e \le 1-\lambda$, then $0 < \tau < \tau_2 \& \frac{1}{2}(3\gamma\tau + \lambda\tau^2) < k < k_2$,
  - (ii) if $0 < \lambda \le \frac{1}{2} \& \lambda < 1-\lambda < e$, then $\tau_1 < \tau < \tau_2 \& \frac{1}{2}(3\gamma\tau + \lambda\tau^2) < k < \min\{k_1, k_2\}$,
  - (iii) if $\frac{1}{2} < \lambda \le \frac{3}{5} \& 1-\lambda < \lambda < e$, then $\tau_1 < \tau < \tau_2 \& \frac{1}{2}(3\gamma\tau + \lambda\tau^2) < k < \min\{k_1, k_2\}$.
- $1-2\theta+2\theta^2 < \lambda < 1-\theta+\theta^2 (0.5 < \lambda < 0.75)$ in the hybrid strategy.
  - (i) $0 < e < \min\{e_1, e_4\} \& \gamma > \gamma_2 \& k > \max\{k_3, k_5\}$,
  - (ii) $e_1 < e < e_4 \& \gamma_2 < \gamma < \gamma_1 \& \max\{k_3, k_5\} < k < k_4$.

The expressions of τ_1, τ_2, τ_3, k_1, k_2, k_3, k_4, k_5, e_1, e_2, e_3, e_4, γ_1 and γ_2 are defined in Appendix B.

Appendix B. Notes

B.1 Values of some variables used in this paper

- Note that $D^{R^*} = \frac{\theta}{2\theta+\lambda-1}$, $D^{IS^*} = \frac{2k(e+\lambda-1)-\tau(\gamma(e+\lambda-1)+e\tau)}{2(1-\lambda)(\tau(2\gamma+\lambda\tau)-2k)}$, and $D^{IT^*} = \frac{2k(1+e-\lambda)-\tau(\gamma(1+e-\lambda)+e\tau)}{2(1-\lambda)(2k-\tau(2\gamma+\lambda\tau))}$.
- Note that $D_R^{HS^*} = \frac{\left[\begin{matrix}\tau(2e\gamma+\gamma(1-\theta)(1-\lambda)+e(1-\theta)(2\theta+\lambda-1)\tau) \\ -k(1-\theta)(e(1+\theta)+(1-\theta)(1+(\theta-1)\theta-\lambda))\end{matrix}\right]}{(1-\theta)^2(2\theta+\lambda-1)(k(\theta-1)+2\gamma\tau)}$ and $D_I^{HS^*} = \frac{\left[\begin{matrix}\tau(\gamma(\theta(3-2e+2(\theta-2)\theta-\lambda)+\lambda-1)+e(\theta-1)(2\theta+\lambda-1)\tau) \\ -k(1-\theta)(\theta(3-2e+2(\theta-2)\theta-\lambda)+\lambda-1)\end{matrix}\right]}{(1-\theta)^2(2\theta+\lambda-1)(k(\theta-1)+2\gamma\tau)}$.
- Note that $D_R^{HT^*} = \frac{\left[\begin{matrix}\tau(e(\theta-1)(2\theta+\lambda-1)\tau-\gamma(2e+\theta+\lambda-\theta\lambda-1)) \\ +k(1-\theta)(e(1+\theta)+(\theta-1)(1+(\theta-1)\theta-\lambda))\end{matrix}\right]}{(1-\theta)^2(2\theta+\lambda-1)(k(\theta-1)+2\gamma\tau)}$ and $D_I^{HT^*} = \frac{\left[\begin{matrix}\tau(\gamma(\theta(3+2e+2(\theta-2)\theta-\lambda)+\lambda-1)+e(1-\theta)(2\theta+\lambda-1)\tau) \\ -k(1-\theta)(\theta(3+2e+2(\theta-2)\theta-\lambda)+\lambda-1)\end{matrix}\right]}{(1-\theta)^2(2\theta+\lambda-1)(k(\theta-1)+2\gamma\tau)}$.

B.2 Expressions used in Appendix A

- Note that

$$\tau_1 = \frac{\left[\begin{array}{l}(1-\lambda)^2(e+\lambda)^2\sqrt{\dfrac{\gamma^2(-1+e+\lambda)^2(e(\lambda-2)+(\lambda-1)\lambda)(5(\lambda-1)\lambda+e(5\lambda-6))}{(1-\lambda)^4(e+\lambda)^4}} \\ +\gamma(-1+e+\lambda)(3(1-\lambda)\lambda+e(4-3\lambda))\end{array}\right]}{2(1+\lambda)^2(e+\lambda)^2},\quad \tau_2 = \frac{1}{2}\left[\begin{array}{l}\dfrac{\gamma(1+e-\lambda)(3(\lambda-1)\lambda+e(4-3\lambda))}{(e-\lambda)^2(-1+\lambda)^2} \\ -\sqrt{\dfrac{\gamma^2(1+e-\lambda)^2\left(e(\lambda-2)+\lambda-\lambda^2\right)(5(1-\lambda)\lambda+e(5\lambda-6))}{(e-\lambda)^4(1-\lambda)^4}}\end{array}\right],$$

$$\tau_3 = \frac{1}{2}\left[\begin{array}{l}\sqrt{\dfrac{\gamma^2(1+e-\lambda)^2\left(e(\lambda-2)+\lambda-\lambda^2\right)(5(1-\lambda)\lambda+e(5\lambda-6))}{(e-\lambda)^4(1-\lambda)^4}} \\ -\dfrac{\gamma(1+e-\lambda)(3(1-\lambda)\lambda-e(4-3\lambda))}{(e-\lambda)^2(1-\lambda)^2}\end{array}\right].$$

- Note that $k_1 = \frac{(\gamma(-1+e+\lambda)+e\tau)^2}{2(-1+e+\lambda)(e(2-\lambda)+(1-\lambda)\lambda)}$, $k_2 = \frac{(\gamma(1+e-\lambda)+e\tau)^2}{2(1+e-\lambda)\left(e(2-\lambda)-\lambda+\lambda^2\right)}$, $k_3 = \frac{\tau(2e\gamma+\gamma(1-\theta)(1-\lambda)+e(1-\theta)(-1+2\theta+\lambda)\tau)}{(1-\theta)(e(1+\theta)+(1-\theta)(1-(1-\theta)\theta-\lambda))}$, $k_4 = \frac{\tau(\gamma-\gamma\lambda+\gamma\theta(2e+2(2-\theta)\theta+\lambda-3)+e(1-\theta)(-1+2\theta+\lambda)\tau)}{(1-\theta)(1-\theta(3-2e-2(2-\theta)\theta-\lambda)+\lambda)}$ and $k_5 = \frac{\tau(\gamma(1-2e-\theta-\lambda+\theta\lambda)-e(1-\theta)(-1+2\theta+\lambda)\tau)}{(\theta-1)(e(1+\theta)+(\theta-1)(1+(\theta-1)\theta-\lambda))}$.
- Note that $e_1 = \frac{(\theta-1)(1+2(\theta-1)\theta-\lambda)}{2\theta}$, $e_2 = \frac{(\theta-1)(1+(\theta-1)\theta-\lambda)}{1+\theta}$, $e_3 = \frac{(1-\theta)(1+2(\theta-1)\theta-\lambda)}{2\theta}$, and $e_4 = \frac{(1-\theta)(1+(\theta-1)\theta-\lambda)}{1+\theta}$.
- Note that $\gamma_1 = \frac{e(1-\theta)(-1+2\theta+\lambda)\tau}{1-\lambda+\theta(2e+2(2-\theta)\theta+\lambda-3)}$ and $\gamma_2 = \frac{e(1-\theta)(-1+2\theta+\lambda)\tau}{\theta(3+2e+2(\theta-2)\theta-\lambda)+\lambda-1}$.

B.3 Expressions used in our model analysis

- Note that $\lambda_p = \frac{\tau-3\gamma}{2\tau}+\frac{1}{2}\sqrt{\frac{5\gamma^2+2\gamma\tau+\tau^2}{\tau^2}}$ and $k_p = \frac{\gamma\left[\begin{array}{l}(1-\theta)\left(4\theta^3+2\theta^2(\lambda-5)+4\theta(1-\lambda)-(1-\lambda)^2\right)\tau \\ +2\gamma\theta(1+2(\theta-1)\theta-\lambda)\end{array}\right]}{(1-\theta)^2\left(4(1-\theta)^2\theta+2\lambda+2(\theta-2)\theta\lambda-\lambda^2-1\right)}$.
- Note that $\Delta = \sqrt{\frac{(h-\theta)^2(\tau(\gamma+\tau)-2k)^2}{\theta(\tau(2\gamma+\lambda\tau)-2k)^2}}$, then $e(\lambda)_1 = \frac{(\lambda-1)\left[\begin{array}{l}4k^2(1+\Delta)-2k\tau(\gamma(2+4\Delta)+\tau+2\Delta\lambda\tau) \\ +\tau^2\left(\gamma^2(1+4\Delta)+\Delta\lambda^2\tau^2+\gamma(\tau+4\Delta\lambda\tau)\right)\end{array}\right]}{(\tau(\gamma+\tau)-2k)^2}$ and $e(\lambda)_2 = \frac{(1-\lambda)\left[\begin{array}{l}4k^2(\Delta-1)-2k\tau(4\gamma\Delta-\tau+2\Delta\lambda\tau-2\gamma) \\ +\tau^2\left(\gamma^2(4\Delta-1)+\gamma(4\Delta\lambda-1)\tau+\Delta\lambda^2\tau^2\right)\end{array}\right]}{(\tau(\gamma+\tau)-2k)^2}$, and $e_5 = \frac{(1-\theta)(k(\theta-1)(1+(\theta-1)\theta-\lambda)+\gamma(1-\lambda)\tau)}{k\left(\theta^2-1\right)+\tau(2\gamma+(1-\theta)(2\theta+\lambda-1)\tau)}$.
- Note that $E_2 = \frac{\left[\begin{array}{l}2k\left(4\theta(1-\lambda)-(1-\lambda)^2+4\theta^3(2-\lambda)\lambda+2\theta^2\lambda((4-\lambda)\lambda-5)\right) \\ -2\theta^2(1+\lambda(2\theta+\lambda-2))(\gamma+\tau)^2\end{array}\right]}{(1-\lambda)^2(2\theta+\lambda-1)^2(\tau(2\gamma+\lambda\tau)-2k)}$, then $e(\lambda)_3 = \frac{(\lambda-1)\left(\gamma(\gamma+\tau)+(1-\lambda)\left((2k-\tau(2\gamma+\lambda\tau))\sqrt{E_2}-2k\right)\right)}{2k(\lambda-2)+(\gamma+\tau)^2}$, $e(\lambda)_4 = \frac{(\lambda-1)\left((1-\lambda)\left((2k-\tau(2\gamma+\lambda\tau))\sqrt{E_2}+2k\right)-\gamma(\gamma+\tau)\right)}{2k(\lambda-2)+(\gamma+\tau)^2}$.
- Note that $E = 4\gamma^2\theta^2-2k(1-\theta)^2\theta(2\theta+\lambda-2)+4\gamma(\theta-1)\theta\tau+(1-\theta)^2(2\theta+\lambda-1)^2\tau^2$ and $E_1 = (\theta-1)^3(k(\theta-1)+2\gamma\tau)\left[\begin{array}{l}2\gamma\theta\left[\begin{array}{l}\gamma(2(1-\theta)\theta+\lambda-1)\left(2\theta(1+2(\theta-2)\theta)+\lambda+2(\theta-2)\theta\lambda-\lambda^2\right) \\ +2(1-\theta)\theta\left(4(1-\theta)^2\theta+2\lambda+2(\theta-2)\theta\lambda-\lambda^2-1\right)\tau\end{array}\right] \\ +k(1-\theta)^2\left(1-4(1-\theta)^2\theta-2\lambda+2(2-\theta)\theta\lambda+\lambda^2\right)^2\end{array}\right]$, then, $e(\lambda)_5 = \frac{\left[\begin{array}{l}\gamma(\theta-1)\left[\begin{array}{l}(\theta-1)\left(4\theta^3+2\theta^2(\lambda-5)+4\theta(1-\lambda)-(1-\lambda)^2\right)\tau \\ +2\gamma\theta(2(1-\theta)\theta+\lambda-1)\end{array}\right] \\ +k(\theta-1)^3\left(4(1-\theta)^2\theta+2\lambda+2(\theta-2)\theta\lambda-\lambda^2-1\right)\end{array}\right]-\sqrt{E_1}}{E}$,

and $e(\lambda)_6 = \frac{\left[\begin{array}{l}\gamma(\theta-1)\left[\begin{array}{l}(\theta-1)\left(4\theta^3+2\theta^2(\lambda-5)+4\theta(1-\lambda)-(1-\lambda)^2\right)\tau \\ +2\gamma\theta(2(1-\theta)\theta+\lambda-1)\end{array}\right] \\ +k(\theta-1)^3\left(4(1-\theta)^2\theta+2\lambda+2(\theta-2)\theta\lambda-\lambda^2-1\right)\end{array}\right]+\sqrt{E_1}}{E}$.

Data availability

Data will be made available on request.

References

Abbas, J., & Khan, S. M. (2023). Green knowledge management and organizational green culture: An interaction for organizational green innovation and green performance. *Journal of Knowledge Management, 27*(7), 1852–1870. https://doi.org/10.1108/jkm-03-2022-0156

Abbas, J., & Sagsan, M. (2019). Impact of knowledge management practices on green innovation and corporate sustainable development: A structural analysis. *Journal of Cleaner Production, 229*, 611–620. https://doi.org/10.1016/j.jclepro.2019.05.024

Abdulmuhsin, A. A., Owain, H. O., & Alkhwaldi, A. F. (2024). Understanding the academic use of KM-driven metaverse technology: Insights from medical colleges. *Journal of Science and Technology Policy Management*. https://doi.org/10.1108/JSTPM-12-2023-0229

Aghamirian, B., Dorri, B., & Aghamirian, B. (2015). Customer knowledge management application in gaining organization's competitive advantage in electronic commerce. *Journal of Theoretical and Applied Electronic Commerce Research, 10*(1), 63–78.

Ahn, S., Jin, B., & Seo, H. (2024). Why do people interact and buy in the metaverse? Self-Expansion perspectives and the impact of hedonic adaptation. *Journal of Business Research, 175*, Article 114557. https://doi.org/10.1016/j.jbusres.2024.114557

Alavi, M., & Leidner, D. E. (2001). Review: Knowledge management and knowledge management systems: Conceptual foundations and research issues. *MIS Quarterly, 25*(1), 107–136. https://doi.org/10.2307/3250961

Alizadeh-Basban, N., & Taleizadeh, A. (2020). A hybrid circular economy - game theoretical approach in a dual-channel green supply chain considering sale's effort, delivery time, and hybrid remanufacturing. *Journal of Cleaner Production, 250*, Article 119521. https://doi.org/10.1016/j.jclepro.2019.119521

Ancillai, C., Sabatini, A., Gatti, M., & Perna, A. (2023). Digital technology and business model innovation: A systematic literature review and future research agenda.

Technological Forecasting and Social Change, 188. https://doi.org/10.1016/j.techfore.2022.122307
Animesh, A., Pinsonneault, A., Yang, S., & Oh, W. (2011). An odyssey into virtual worlds: Exploring the impacts of technological and spatial environments on intention to purchase virtual products. *MIS Quarterly, 35*(3), 789–810.
Arani, H. V., Rabbani, M., & Rafiei, H. (2016). A revenue-sharing option contract toward coordination of supply chains. *International Journal of Production Economics, 178*, 42–56. https://doi.org/10.1016/j.ijpe.2016.05.001
Avinadav, T., Chernonog, T., Meilijson, I., & Perlman, Y. (2022). A consignment contract with revenue sharing between an app developer and a distribution platform. *International Journal of Production Economics, 243*, Article 108322. https://doi.org/10.1016/j.ijpe.2021.108322
Bag, S., Gupta, S., Kumar, A., & Sivarajah, U. (2021). An integrated artificial intelligence framework for knowledge creation and B2B marketing rational decision making for improving firm performance. *Industrial Marketing Management, 92*, 178–189. https://doi.org/10.1016/j.indmarman.2020.12.001
Barham, H., Dabic, M., Daim, T., & Shifrer, D. (2020). The role of management support for the implementation of open innovation practices in firms. *Technology in Society, 63*. https://doi.org/10.1016/j.techsoc.2020.101282
Bresciani, S., Rehman, S. U., Giovando, G., & Alam, G. M. (2023). The role of environmental management accounting and environmental knowledge management practices influence on environmental performance: Mediated-moderated model. *Journal of Knowledge Management, 27*(4), 896–918. https://doi.org/10.1108/jkm-12-2021-0953
Bruni, R., Piccarozzi, M., Caboni, F., & Practice. (2025). Defining the Metaverse with challenges and opportunities in the business environment. *Journal of Marketing Theory, 33*(1), 60–77. https://doi.org/10.1080/10696679.2023.2273555
Buechel, H., & Spinler, S. (2024). The impact of the metaverse on e-commerce business models - a delphi-based scenario study. *Technology in Society, 76*. https://doi.org/10.1016/j.techsoc.2024.102465
Cai, J., Dong, R., Zeng, Z., & Hu, X. (2022). Supply chain competition models with strategic customers considering sales effort. *Computers & Industrial Engineering, 172*, Article 108566. https://doi.org/10.1016/j.cie.2022.108566
Centobelli, P., & Ndou, V. (2019). Managing customer knowledge through the use of big data analytics in tourism research. *Current Issues in Tourism, 22*(15), 1862–1882. https://doi.org/10.1080/13683500.2018.1564739
Chakraborty, D., Polisetty, A., & Rana, N. (2024). Consumers' continuance intention towards metaverse-based virtual stores: A multi-study perspective. *Technological Forecasting and Social Change, 203*, Article 123405. https://doi.org/10.1016/j.techfore.2024.123405
Chang, Y. C. (2003). Benefits of co-operation on innovative performance: Evidence from integrated circuits and biotechnology firms in the UK and Taiwan. *R&D Management, 33*(4), 425–437. https://doi.org/10.1111/1467-9310.00308
Chen, P.-K., & Huang, X. (2024). Enhancing supply chain resilience and realizing green sustainable development through the virtual environment of the metaverse. *Sustainable Development (Bradford), 32*(1), 438–454. https://doi.org/10.1002/sd.2663
Chen, X., Wang, Y., & Wei, S. (2023). How does social media influence car sales: The effects of firm-, user-, and professional-generated content. *Information Technology and Management*. https://doi.org/10.1007/s10799-023-00414-x. ahead-of-print.
Chen, P.-K., Ye, Y., & Huang, X. (2023). The metaverse in supply chain knowledge sharing and resilience contexts: An empirical investigation of factors affecting adoption and acceptance. *Journal of Innovation & Knowledge, 8*(4). https://doi.org/10.1016/j.jik.2023.100446
Cresnar, R., Dabic, M., Stojcic, N., & Nedelko, Z. (2023). It takes two to tango: Technological and non-technological factors of industry 4.0 implementation in manufacturing firms. *Review of Managerial Science, 17*(3), 827–853. https://doi.org/10.1007/s11846-022-00543-7
Di Vaio, A., Palladino, R., Pezzi, A., & Kalisz, D. E. (2021). The role of digital innovation in knowledge management systems: A systematic literature review. *Journal of Business Research, 123*, 220–231. https://doi.org/10.1016/j.jbusres.2020.09.042
Dolgui, A., & Ivanov, D. (2022). 5G in digital supply chain and operations management: Fostering flexibility, end-to-end connectivity and real-time visibility through internet-of-everything. *International Journal of Production Research, 60*(2), 442–451. https://doi.org/10.1080/00207543.2021.2002969
Gao, F., & Su, X. (2017). Online and offline information for omnichannel retailing. *M&SOM-Manufacturing & Service Operations Management, 19*(1), 84–98. https://doi.org/10.1287/msom.2016.0593
Ghadimi, S., Szidarovszky, F., Farahani, R. Z., & Khiabani, A. Y. (2013). Coordination of advertising in supply chain management with cooperating manufacturer and retailers. *IMA Journal of Management Mathematics, 24*(1), 1–19. https://doi.org/10.1093/imaman/dpr022
Giraldo, S. M., Aguilar, L. J., Giraldo, L. M., & Toro, I. D. (2019). Techniques for the identification of organizational knowledge management requirements. *Journal of Knowledge Management, 23*(7), 1355–1402. https://doi.org/10.1108/jkm-08-2018-0479
Grant, R. M. (1996). Toward a knowledge-based theory of the firm. *Strategic Management Journal, 17*(S2), 109–122. https://doi.org/10.1002/smj.4250171110
Gu, M., Yang, L., & Huo, B. (2021). The impact of information technology usage on supply chain resilience and performance: An ambidexterous view. *International Journal of Production Economics, 232*, Article 107956. https://doi.org/10.1016/j.ijpe.2020.107956
Gulati, R. (1998). Alliances and networks. *Strategic Management Journal, 19*(4), 293–317. https://doi.org/10.1002/(SICI)1097-0266(199804)19:4<293::AID-SMJ982>3.0.CO;2-M
Gursoy, D., Malodia, S., & Dhir, A. (2022). The metaverse in the hospitality and tourism industry: An overview of current trends and future research directions. *Journal of Hospitality Marketing & Management, 31*(5), 527–534. https://doi.org/10.1080/19368623.2022.2072504
Hadi, R., Melumad, S., & Park, E. S. (2024). The metaverse: A new digital frontier for consumer behavior. *Journal of Consumer Psychology, 34*(1), 142–166. https://doi.org/10.1002/jcpy.1356
Hoyer, W., Kroschke, M., Schmitt, B., Kraume, K., & Shankar, V. (2020). Transforming the customer experience through new technologies. *Journal of Interactive Marketing, 51*, 57–71. https://doi.org/10.1016/j.intmar.2020.04.001
Huang, S., Gao, Y., & Wang, Y. (2023). Supply chain channel structure with online platform and dual-purpose firms. *Transportation Research Part E: Logistics and Transportation Review, 180*. https://doi.org/10.1016/j.tre.2023.103363
Huang, F., He, J., & Lei, Q. (2020). Coordination in a retailer-dominated supply chain with a risk-averse manufacturer under marketing dependency. *International Transactions in Operational Research, 27*(6), 3056–3078. https://doi.org/10.1111/itor.12520
Huynh-The, T., Pham, Q.-V., Pham, X.-Q., Nguyen, T. T., Han, Z., & Kim, D.-S. (2023). Artificial intelligence for the metaverse: A survey. *Engineering Applications of Artificial Intelligence, 117*. https://doi.org/10.1016/j.engappai.2022.105581
Hwang, E. H., & Krackhardt, D. (2020). Online knowledge communities: Breaking or sustaining knowledge silos? *Production and Operations Management, 29*(1), 138–155. https://doi.org/10.1111/poms.13098
Im, H., & Lee, G. (2023). Do consumers always believe humans create better boxes than AI? The context-dependent role of recommender creativity. *International Journal of Retail & Distribution Management, 51*(8), 1045–1060. https://doi.org/10.1108/ijrdm-09-2021-0449
Jafar, R., Ahmad, W., & Sun, Y. (2023). Unfolding the impacts of metaverse aspects on telepresence, product knowledge, and purchase intentions in the metaverse stores. *Technology in Society, 74*, Article 102265. https://doi.org/10.1016/j.techsoc.2023.102265
Jang, S., & Nemeh, A. (2017). Salespeople knowledge search behavior and sales performance: An investigation of printing equipment industry. *Technological Forecasting and Social Change, 120*, 323–333. https://doi.org/10.1016/j.techfore.2016.12.014
Jen, C. T., Hu, J., Zheng, J., & Xiao, L. L. (2020). The impacts of corporate governance mechanisms on knowledge sharing and supply chain performance. *International Journal of Logistics Research and Applications, 23*(4), 337–353. https://doi.org/10.1080/13675567.2019.1691515
Krotov, V. (2017). The Internet of Things and new business opportunities. *Business Horizons, 60*(6), 831–841. https://doi.org/10.1016/j.bushor.2017.07.009
Kshetri, N. (2022). Web 3.0 and the metaverse shaping organizations' brand and product strategies. *IT Professional, 24*(2), 11–15. https://doi.org/10.1109/mitp.2022.3157206
Kuo, H.-T., & Choi, T.-M. (2024). Metaverse in transportation and logistics operations: An AI-supported digital technological framework. *Transportation Research Part E: Logistics and Transportation Review, 185*. https://doi.org/10.1016/j.tre.2024.103496
Lemon, K., & Verhoef, P. (2016). Understanding customer experience throughout the customer journey. *Journal of Marketing, 80*(6), 69–96. https://doi.org/10.1509/jm.15.0420
Li, G., Zhang, T., & Tayi, G. K. (2020). Inroad into omni-channel retailing: Physical showroom deployment of an online retailer. *European Journal of Operational Research, 283*(2), 676–691. https://doi.org/10.1016/j.ejor.2019.11.032
Liu, P. L., Chen, W. C., & Tsai, C. H. (2004). An empirical study on the correlation between knowledge management capability and competitiveness in Taiwan's industries. *Technovation, 24*(12), 971–977. https://doi.org/10.1016/S0166-4972(03)00061-0
Liu, P., Long, Y., Song, H.-C., & He, Y.-D. (2020). Investment decision and coordination of green agri-food supply chain considering information service based on blockchain and big data. *Journal of Cleaner Production, 277*. https://doi.org/10.1016/j.jclepro.2020.123646
Liu, Y., Ying, Z., Ying, Y., Wang, D., & Chen, J. (2024). Artificial intelligence orientation and internationalization speed: A knowledge management perspective. *Technological Forecasting and Social Change, 205*. https://doi.org/10.1016/j.techfore.2024.123517
Loorbach, D., & Wijsman, K. (2013). Business transition management: Exploring a new role for business in sustainability transitions. *Journal of Cleaner Production, 45*, 20–28. https://doi.org/10.1016/j.jclepro.2012.11.002
Mancuso, I., Petruzzelli, A. M., Panniello, U., & Nespoli, C. (2024). A microfoundation perspective on business model innovation: The cases of Roblox and meta in metaverse. *IEEE Transactions on Engineering Management, 71*, 12750–12763. https://doi.org/10.1109/tem.2023.3275198
Martins, V. W. B., Rampasso, I. S., Anholon, R., Quelhas, O. L. G., & Filho, W. L. (2019). Knowledge management in the context of sustainability: Literature review and opportunities for future research. *Journal of Cleaner Production, 229*, 489–500. https://doi.org/10.1016/j.jclepro.2019.04.354
Modgil, S., Gupta, S., Stekelorum, R., & Laguir, I. (2022). AI technologies and their impact on supply chain resilience during COVID-19. *International Journal of Physical Distribution & Logistics Management, 52*(2), 130–149. https://doi.org/10.1108/ijpdlm-12-2020-0434
Modgil, S., Singh, R., & Hannibal, C. (2022). Artificial intelligence for supply chain resilience: Learning from COVID-19. *International Journal of Logistics Management, 33*(4), 1246–1268. https://doi.org/10.1108/ijlm-02-2021-0094
Moro-Visconti, R. (2022). From physical reality to the metaverse: A multilayer network valuation. *Journal of Metaverse, 2*(1), 16–22.
Mothe, C., Nguyen-Thi, U. T., & Triguero, A. (2018). Innovative products and services with environmental benefits: Design of search strategies for external knowledge and

absorptive capacity. *Journal of Environmental Planning and Management, 61*(11), 1934–1954. https://doi.org/10.1080/09640568.2017.1372275
Okhuysen, G. A., & Eisenhardt, K. M. (2002). Integrating knowledge in groups: How formal interventions enable flexibility. *Organization Science, 13*(4), 370–386. https://doi.org/10.1287/orsc.13.4.370.2947
Olan, F., Suklan, J., Arakpogun, E. O., & Robson, A. (2024). Advancing consumer behavior: The role of artificial intelligence technologies and knowledge sharing. *IEEE Transactions on Engineering Management, 71*, 13227–13239. https://doi.org/10.1109/tem.2021.3083536
Philsoophian, M., Akhavan, P., & Namvar, M. (2022). The mediating role of blockchain technology in improvement of knowledge sharing for supply chain management. *Management Decision, 60*(3), 784–805. https://doi.org/10.1108/md-08-2020-1122
Pillai, R., Sivathanu, B., Rana, N., Preet, R., & Mishra, A. (2023). Factors influencing customers' apparel shopping intention in metaverse. *Journal of Computer Information Systems*. https://doi.org/10.1080/08874417.2023.2280842. ahead-of-print.
Qi, Y., Wang, X., Zhang, M., & Wang, Q. (2023). Developing supply chain resilience through integration: An empirical study on an e-commerce platform. *Journal of Operations Management, 69*(3), 477–496. https://doi.org/10.1002/joom.1226
Ramon Saura, J., Palacios-Marques, D., & Barbosa, B. (2023). A review of digital family businesses: Setting marketing strategies, business models and technology applications. *International Journal of Entrepreneurial Behavior & Research, 29*(1), 144–165. https://doi.org/10.1108/ijebr-03-2022-0228
Reagans, R., & McEvily, B. (2003). Network structure and knowledge transfer: The effects of cohesion and range. *Administrative Science Quarterly, 48*(2), 240–267. https://doi.org/10.2307/3556658
Schniederjans, D. G., Curado, C., & Khalajhedayati, M. (2020). Supply chain digitisation trends: An integration of knowledge management. *International Journal of Production Economics, 220*. https://doi.org/10.1016/j.ijpe.2019.07.012
Scuotto, V., Lemaire, S. L. L., Magni, D., & Maalaoui, A. (2022). Extending knowledge-based view: Future trends of corporate social entrepreneurship to fight the gig economy challenges. *Journal of Business Research, 139*, 1111–1122. https://doi.org/10.1016/j.jbusres.2021.10.060
Shen, B., Zhu, C., Li, Q., & Wang, X. (2021). Green technology adoption in textiles and apparel supply chains with environmental taxes. *International Journal of Production Research, 59*(14), 4157–4174. https://doi.org/10.1080/00207543.2020.1758354
Sjodin, D., Parida, V., Jovanovic, M., & Visnjic, I. (2020). Value creation and value capture alignment in business model innovation: A process view on outcome-based business models. *Journal of Product Innovation Management, 37*(2), 158–183. https://doi.org/10.1111/jpim.12516
snobette.com. (2021). Nike partners with gaming platform Roblox with launch of Nikeland. Retrieved from https://snobette.com/2021/11/nike-debuts-nikeland-with-online-gaming-platform-roblox/. (Accessed 7 February 2024).
Song, B., Zhang, M., & Wu, P. (2022). Driven by technology or sociality? Use intention of service robots in hospitality from the human-robot interaction perspective. *International Journal of Hospitality Management, 106*, Article 103278. https://doi.org/10.1016/j.ijhm.2022.103278
Sun, M. (2012). How does the variance of product ratings matter? *Management Science, 58*(4), 696–707. https://doi.org/10.1287/mnsc.1110.1458
Sun, C., & Ji, Y. (2022). For better or for worse: Impacts of IoT technology in e-commerce channel. *Production and Operations Management, 31*(3), 1353–1371. https://doi.org/10.1111/poms.13615
Sun, Y. H., Wang, Z. H., Yan, S., & Han, X. H. (2022). Digital showroom strategies for dual-channel supply chains in the presence of consumer webrooming behavior. *Annals of Operations Research*. https://doi.org/10.1007/s10479-021-04475-5
Taleizadeh, A. A., Soleymanfar, V. R., & Choi, T.-M. (2017). Optimal pricing and alliance strategy in a retailer-led supply chain with the return policy: A game-theoretic analysis. *Information Sciences, 420*, 466–489. https://doi.org/10.1016/j.ins.2017.08.054
Tomini, N., Brenk, K., Kanbach, D. K., Stubner, S., & Bernhard, F. (2024). From hype to impact: How companies can leverage the metaverse technology for business models. *The International Journal of Entrepreneurship and Innovation*. , Article 14657503241258934. https://doi.org/10.1177/14657503241258934
Tong, Y., & Xiao, T. (2024). National or third-party manufacturer? Sourcing strategy of a dominant platform: Signaling game's perspective. *Omega-International Journal of Management Science, 124*. https://doi.org/10.1016/j.omega.2023.103016
Tsunoda, Y., & Zennyo, Y. (2021). Platform information transparency and effects on third-party suppliers and offline retailers. *Production and Operations Management, 30*(11), 4219–4235. https://doi.org/10.1111/poms.13518
Ul-Durar, S., Awan, U., Varma, A., Memon, S., & Mention, A.-L. (2023). Integrating knowledge management and orientation dynamics for organization transition from eco-innovation to circular economy. *Journal of Knowledge Management, 27*(8), 2217–2248. https://doi.org/10.1108/jkm-05-2022-0424
von Delft, S., Kortmann, S., Gelhard, C., & Pisani, N. (2019). Leveraging global sources of knowledge for business model innovation. *Long Range Planning, 52*(5). https://doi.org/10.1016/j.lrp.2018.08.003
Wang, W.-L., Malthouse, E. C., Calder, B., & Uzunoglu, E. (2019). B2B content marketing for professional services: In-person versus digital contacts. *Industrial Marketing Management, 81*, 160–168. https://doi.org/10.1016/j.indmarman.2017.11.006
Wang, M., Yu, H., Bell, Z., & Chu, X. (2022). Constructing an edu-metaverse ecosystem: A new and innovative framework. *IEEE Transactions on Learning Technologies, 15*(6), 685–696. https://doi.org/10.1109/tlt.2022.3210828
Wiig, K. M. (1997). Integrating intellectual capital and knowledge management. *Long Range Planning, 30*(3), 399–405. https://doi.org/10.1016/S0024-6301(97)90256-9
Wijethilake, C. (2017). Proactive sustainability strategy and corporate sustainability performance: The mediating effect of sustainability control systems. *Journal of Environmental Management, 196*, 569–582. https://doi.org/10.1016/j.jenvman.2017.03.057
Wu, Z. Y., Ming, X. G., Wang, Y. L., & Wang, L. (2014). Technology solutions for product lifecycle knowledge management: Framework and a case study. *International Journal of Production Research, 52*(21), 6314–6334. https://doi.org/10.1080/00207543.2014.935823
Wu, J., & Wang, X. (2023). Platform-leading blockchain adoption for traceability under upstream competition. *Annals of Operations Research*. https://doi.org/10.1007/s10479-022-05147-8
Xin, B., Song, Y., Tan, H., & Peng, W. (2025). Sustainable digital fashion in a metaverse ecosystem. *Journal of Retailing and Consumer Services, 82*. https://doi.org/10.1016/j.jretconser.2024.104099
Xin, B., Song, Y., & Xie, L. (2024). Dynamic pricing and service customization strategy for IoT-based smart products. *Technological Forecasting and Social Change, 199*, Article 123046. https://doi.org/10.1016/j.techfore.2023.123046
Xu, X., Hong, Z., Chen, Y., & Cheng, T. (2023). When is it wise to use artificial intelligence for platform operations considering consumer returns? *European Journal of Operational Research, 308*(3), 1188–1205. https://doi.org/10.1016/j.ejor.2022.11.036
Yan, X., & Han, X. (2022). Optimal pricing and remanufacturing entry strategies of manufacturers in the presence of online reviews. *Annals of Operations Research, 316*, 59–92. https://doi.org/10.1007/s10479-021-04030-2
Yan, M.-R., Hong, L.-Y., & Warren, K. (2022). Integrated knowledge visualization and the enterprise digital twin system for supporting strategic management decision. *Management Decision, 60*(4), 1095–1115. https://doi.org/10.1108/md-02-2021-0182
Yang, Y., Gong, Y., Land, L., & Chesney, T. (2020). Understanding the effects of physical experience and information integration on consumer use of online to offline commerce. *International Journal of Information Management, 51*, Article 102046. https://doi.org/10.1016/j.ijinfomgt.2019.102046
Yang, W., Zhang, J., & Yan, H. (2021). Impacts of online consumer reviews on a dual-channel supply chain. *Omega-International Journal of Management Science, 101*, Article 102266. https://doi.org/10.1016/j.omega.2020.102266
Yi, Y., Wang, Y., & Shu, C. (2020). Business model innovations in China: A focus on value propositions. *Business Horizons, 63*(6), 787–799. https://doi.org/10.1016/j.bushor.2020.07.002
Yoo, K., Welden, R., Hewett, K., & Haenlein, M. (2023). The merchants of meta: A research agenda to understand the future of retailing in the metaverse. *Journal of Retailing, 99*(2), 173–192. https://doi.org/10.1016/j.jretai.2023.02.002
Zabel, C., O'Brien, D., & Natzel, J. (2023). Sensing the Metaverse: The microfoundations of complementor firms' dynamic sensing capabilities in emerging-technology ecosystems. *Technological Forecasting and Social Change, 192*, Article 122562. https://doi.org/10.1016/j.techfore.2023.122562
Zahoor, N., Khan, Z., Marinova, S., & Cui, L. (2024). Ambidexterity in strategic alliances: An integrative review of the literature. *International Journal of Management Reviews, 26*(1), 82–109. https://doi.org/10.1111/ijmr.12348
Zhang, Y., He, Y., Yue, J., & Gou, Q. (2019). Pricing decisions for a supply chain with refurbished products. *International Journal of Production Research, 57*(9), 2867–2900. https://doi.org/10.1080/00207543.2018.1543968
Zhang, S., Wei, L., & Zhang, J. (2022). Demand forecast sharing for a dominant retailer with supplier encroachment and quality decisions. *European Journal of Operational Research, 301*(1), 39–50. https://doi.org/10.1016/j.ejor.2021.09.024
Zhang, Z., Xu, H., Ke, G. Y., & Chen, K. (2022). Selecting online distribution modes for differentiated products in a platform supply chain. *International Journal of Production Economics, 244*. https://doi.org/10.1016/j.ijpe.2021.108384
Zhang, Q., & Ye, Q. (2023). R2V platform's business model reconstruction in the metaverse era: Based on network effects and bundling. *Information Processing & Management, 60*(6). https://doi.org/10.1016/j.ipm.2023.103503
Zhou, Y.-W., Cao, Z.-H., & Zhong, Y. (2015). Pricing and alliance selection for a dominant retailer with an upstream entry. *European Journal of Operational Research, 243*(1), 211–223. https://doi.org/10.1016/j.ejor.2014.11.004
Zott, C., & Amit, R. (2008). The fit between product market strategy and business model: Implications for firm performance. *Strategic Management Journal, 29*(1), 1–26. https://doi.org/10.1002/smj.642
Zott, C., Amit, R., & Massa, L. (2011). The business model: Recent developments and future research. *Journal of management, 37*(4), 1019–1042. https://doi.org/10.1177/0149206311406265